MIL-STD-1764
31 JANUARY 1980

MILITARY STANDARD

WASHERS
PREFERRED FOR DESIGN,
LISTING OF

FSC 5310

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 304
WASHER, FLAT, SQUARE WITH RADIUS
APPLICABLE DOCUMENT: NAS1401

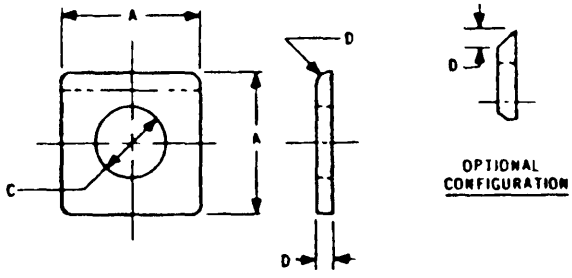


TABLE I. Part numbers.

Material			Aluminum alloy	Carbon steel	CRES
Protective finish			Anodize	Cadmium plate	Passivate
C Min	A Max	D	NAS1401 - Dash no.		
.193	.490	.090 .160	-3D3 -3D5	-3-3 -3-5	-3C3 -3C5
.256	.570	.090 .160	-4D3 -4D5	-4-3 -4-5	-4C3 -4C5
.318	.640	.090 .160	-5D3 -5D5	-5-3 -5-5	-5C3 -5C5
.380	.710	.090 .160 .250	-6D3 -6D5 -6D8	-6-3 -6-5 -6-8	-6C3 -6C5 -6C8
.443	.780	.090 .160 .250	-7D3 -7D5 -7D8	-7-3 -7-5 -7-8	-7C3 -7C5 -7C8
.505	.930	.090 .160 .250	-8D3 -8D5 -8D8	-8-3 -8-5 -8-8	-8C3 -8C5 -8C8
.630	1.150	.090 .160 .250	-10D3 -10D5 -10D8	-10-3 -10-5 -10-8	-10C3 -10C5 -10C8
.755	1.290	.090 .160 .250	-12D3 -12D5 -12D8	-12-3 -12-5 -12-8	-12C3 -12C5 -12C8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 401
WASHER KEY, BEARING
APPLICABLE DOCUMENTS MS19069, MS19070, MS17277 thru MS17328

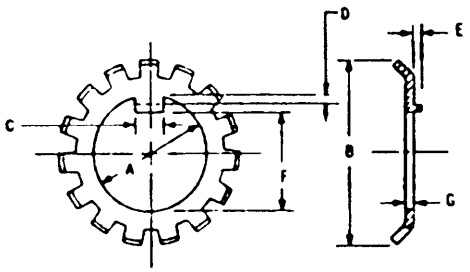


TABLE I. Part numbers.

Material							Steel	Carbon steel	Carbon steel	CRES	Alloy steel
Protective finish							Phosphate	None	Zinc or Cd plate	Passivate	Phosphate
A Min	B Max	C Max	D Max	E Min	F Min	G Nom	Part number	MS19070 - Dash no.			MS19069 - Dash no.
.406	.828	.120	.030	.062	--	.042	--	--	--	--	-A00
.406	.875	.120	.030	.062	--	.042	--	-001	-002	-003	--
.484	.922	.120	.030	.062	--	.042	--	--	--	--	-A01
.484	1.016	.120	.030	.062	--	.042	--	-011	-012	-013	--
.515	1.156	.120	--	.062	.443	.040	MS17277I	--	--	--	--
.601	1.094	.120	.030	.062	--	.042	--	--	--	--	-A02
.601	1.156	.120	.030	.062	--	.042	--	-021	-022	-023	--
.641	1.312	.120	--	.062	.567	.040	MS17277J	--	--	--	--
.679	1.203	.120	.030	.062	--	.042	--	--	--	--	-A03
.679	1.328	.120	.030	.062	--	.042	--	-031	-032	-033	--
.770	1.578	.176	--	.062	.677	.040	MS17277S	--	--	--	--
.801	1.344	.176	.030	.062	--	.042	--	--	--	--	-A04
.801	1.531	.176	.030	.062	--	.042	--	-041	-042	-043	--
.895	1.719	.176	--	.094	.802	.050	MS17277T	--	--	--	--
.989	1.562	.176	.030	.094	--	.050	--	--	--	--	-A05
.989	1.719	.176	.030	.094	--	.050	--	-051	-052	-053	--
1.020	1.891	.176	--	.094	.927	.050	MS17277U	--	--	--	--
1.145	2.047	.176	--	.094	1.052	.050	MS17277V	--	--	--	--
1.193	1.859	.176	.050	.094	--	.050	--	--	--	--	-A06
1.193	1.922	.176	.050	.094	--	.050	--	-061	-062	-063	--
1.270	2.172	.176	--	.094	1.177	.050	MS17277W	--	--	--	--
1.333	2.750	.176	.050	.094	--	.050	--	-0651	-0652	-0653	--
1.395	2.297	.176	--	.094	1.302	.050	MS17277X	--	--	--	--
1.396	2.078	.176	.050	.094	--	.050	--	--	--	--	-A07
1.396	2.250	.176	.050	.094	--	.050	--	-071	-072	-073	--
1.520	2.406	.239	--	.094	1.411	.063	MS17277Y	--	--	--	--
1.583	2.250	.300	.050	.094	--	.058	--	--	--	--	-A08
1.583	2.469	.290	.050	.094	--	.058	--	-081	-082	-083	--
1.775	2.719	.239	--	.125	1.661	.063	MS17277Z	--	--	--	--
1.792	2.500	.300	.050	.125	--	.058	--	--	--	--	-A09
1.792	2.734	.290	.050	.125	--	.058	--	-091	-092	-093	--
1.992	2.688	.300	.050	.125	--	.058	--	--	--	--	-A10
1.992	2.922	.290	.050	.125	--	.058	--	-101	-102	-103	--
2.025	2.969	.239	--	.125	1.911	.063	MS17278A	--	--	--	--
2.182	2.953	.300	.050	.125	--	.063	--	--	--	--	-A11
2.182	3.109	.290	.050	.125	--	.063	--	-111	-112	-113	--
2.281	3.234	.239	--	.125	2.161	.063	MS17278B	--	--	--	--
2.400	3.188	.300	.070	.125	--	.063	--	--	--	--	-A12
2.400	3.344	.290	.070	.125	--	.063	--	-121	-122	-123	--
2.531	3.484	.239	--	.125	2.411	.063	MS17278C	--	--	--	--
2.588	3.375	.300	.070	.125	--	.063	--	--	--	--	-A13
2.588	3.578	.290	.070	.125	--	.063	--	-131	-132	-133	--

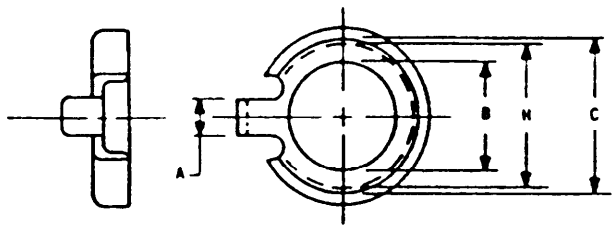
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE I Part numbers. - Continued

Material							Steel	Carbon steel	Carbon steel	CRES	Alloy steel
Protective finish							Phosphate	None	Zinc or cd plate	Passivate	Phosphate
A Min	B Max	C Max	D Max	E Min	F Min	G Nom	Part number	MS19070 - Dash no.			MS19069 - Dash no.
2.701	3.594	.300	.070	.186	--	.063	—	—	—	--	-A14
2.791	3.875	.290	.070	.188	--	.063	—	-141	-142	-143	--
2.913	3.911	.300	.070	.188	--	.063	—	--	--	--	-A15
2.972	4.109	.290	.070	.188	--	.072	--	-151	-152	-153	—
3.177	4.141	.360	.070	.168	--	.053	--	--	--	--	-A16
3.178	4.375	.359	.070	.185	--	.072	--	-161	-162	-163	--
3.356	4.359	.300	.070	.168	--	.360	--	--	--	--	-A17
3.356	4.625	.359	.070	.188	--	.072	—	-171	-172	-173	--
3.522	4.547	.360	.070	.188	--	.063	—	--	--	--	-A18
3.522	4.938	.353	.070	.168	--	.094	—	-181	-182	-183	--
3.800	4.812	.360	.070	.188	--	.063	—	—	—	—	-A19
3.600	5.219	.353	.070	.188	—	.094	—	-191	-192	-193	—
3.988	5.500	.360	.085	.250	--	.063	—	--	—	—	-A20
3.986	5.000	.353	.085	.250	—	.094	--	-201	-202	-203	--
4.192	5.188	.360	.085	.250	--	.063	--	—	—	--	-A21
4.192	5.703	.353	.085	.250	—	.094	--	-211	-212	-213	—
4.395	5.406	.360	.085	.250	—	.063	—	—	—	--	-A22
4.355	6.062	.353	.085	.250	--	.125	—	-221	-222	-223	—
4.801	6.469	.353	.085	.250	--	.125	—	-241	-242	-243	—
5.191	7.031	.435	.105	.250	--	.125	—	-261	-262	-263	—
5.582	7.438	.920	.105	.250	--	.125	—	-281	-282	-283	—
5.983	8.062	.590	.105	.312	--	.156	—	-301	-302	-303	--
6.389	8.438	.590	.105	.312	--	.156	—	-321	-322	-323	—
6.764	9.062	.715	.105	.312	—	.156	—	-341	-342	-343	—
7.171	9.438	.715	.105	.312	—	.156	--	-361	-362	-363	—
7.577	9.875	.715	.105	.312	--	.156	—	-381	-382	-383	—
7.982	10.312	.840	.105	.312	—	.156	—	-401	-402	-403	—

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 402
WASHER, KEY, CUP, LOCK
APPLICABLE DOCUMENT MS9080



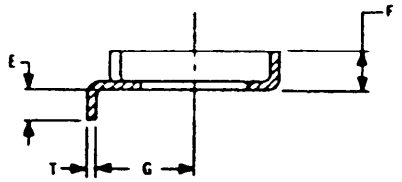


TABLE I Part numbers.

Material	CRES
Protective finish	None

B Min	A Max	C Max	E Min	F Max	G Min	T Max	H	MS9080 Dash no.
.230	.110	.377	.095	.140	.226	.025	.347	-09
.290	.110	.440	.100	.165	.270	.036	.410	-10
.352	.140	.502	.100	.165	.330	.036	.472	-11
.415	.140	.564	.140	.165	.370	.036	.534	-12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 403
WASHER, KEY, CUP, LOCK, SPANNER NUT
APPLICABLE DOCUMENT MS9952

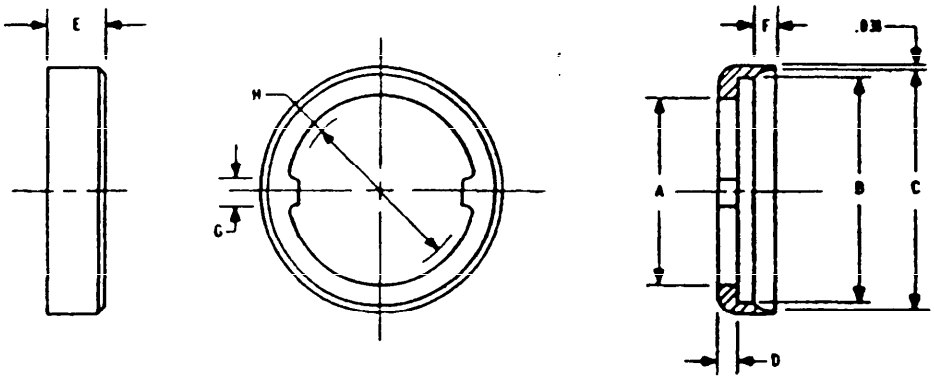


TABLE I. Part numbers.

Material								CRES
Protective finish								None
Hardness (BHN)								140-220
A Min	B Min	C	D	E	F	G Max	H Min	MS9952 + Dash no.
.378	.600	.615	.125	.425	.200	.120	.2756	-02
.441	.750	.775	.125	.425	.200	.120	.3346	-03
.503	.800	.825	.125	.425	.200	.120	.3906	-04
.628	.925	.950	.125	.425	.200	.120	.5156	-06
.753	1.075	1.100	.125	.425	.200	.120	.6406	-08
.878	1.200	1.225	.125	.425	.200	.150	.7656	-10
1.003	1.350	1.375	.125	.425	.200	.150	.8906	-12
1.128	1.475	1.500	.150	.475	.225	.150	1.0156	-14
1.253	1.675	1.650	.150	.475	.225	.180	1.1130	-16
1.378	1.750	1.775	.150	.475	.225	.180	1.2500	-18
1.503	1.900	1.925	.150	.475	.225	.180	1.375	-20
1.753	2.175	2.200	.150	.475	.225	.245	1.630	-24
2.003	2.450	2.475	.150	.475	.225	.245	1.865	-28
2.253	2.700	2.750	.150	.525	.275	.245	2.115	-32
2.503	2.975	3.025	.150	.525	.275	.245	2.365	-36
2.753	3.250	3.300	.150	.525	.275	.245	2.590	-39
3.003	3.525	3.575	.150	.525	.275	.245	2.835	-41

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 404
WASHER, KEY, DRIVE (SYNCHRO)
APPLICABLE DOCUMENT: MS17186

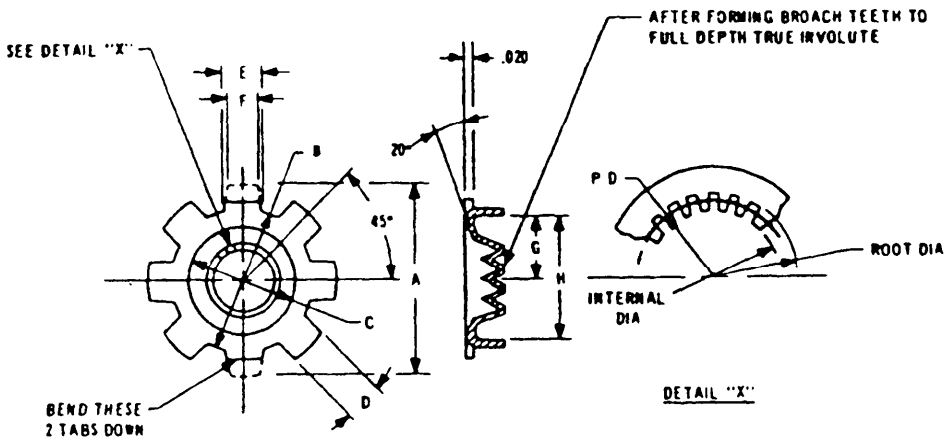


TABLE I. Part numbers.

Material														Aluminum alloy	Brass
Protective finish														Dichromate	Black nickel plating
C Min	A	B	D Min	E	F	G	H	Gear data: Press angle	Gear data: Number of teeth	Gear data: D. P.	Gear data: P. D. Min	Gear data: Root Min	Gear data: Internal Min	MS17186 - Dash no.	
.215	.432	.287	.09	.093	.078	.146	.292	20°	13	120	.108	.125	.091	-1	-5
.291	.516	.350	.09	.125	.093	.177	.355	20°	15	96	.156	.177	.135	-3	-7
									21	120	.175	.192	.158	-2	-6
.350	.625	.437	.12	.125	.093	.208	.417	20°	22	96	.229	.250	.208	-4	-8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 405
WASHER, KEY, FLAT, GENERAL PURPOSE
APPLICABLE DOCUMENT MS27111

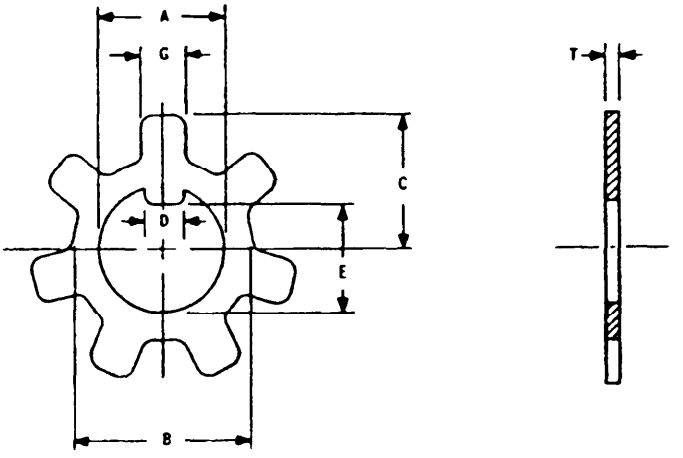


TABLE I. Part numbers.

Material							Carbon steel
Protective finish							Cadmium plate
A	B	C	D	E	G	T Max	MS27111 + Dash no.
.256	.374	.317	.055	.223	.094	.033	-1
.319	.436	.364	.055	.286	.116	.033	-2
.381	.499	.411	.086	.333	.140	.033	-3
.444	.560	.458	.086	.396	.164	.033	-4
.506	.684	.546	.117	.442	.188	.033	-5
.631	.896	.705	.148	.551	.234	.052	-7
.756	1.019	.798	.180	.661	.280	.052	-8
.881	1.206	.934	.180	.786	.328	.052	-9
1.006	1.367	1.052	.242	.880	.374	.065	-10
1.131	1.517	1.211	.242	1.005	.422	.065	-11
1.256	1.698	1.361	.305	1.097	.468	.065	-12
1.381	1.880	1.487	.305	1.222	.516	.065	-13
1.506	2.029	1.623	.367	1.316	.562	.081	-14
1.756	2.392	1.875	.430	1.534	.656	.081	-15
2.006	2.754	2.127	.492	1.753	.750	.081	-16

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 406
WASHER, KEY, LOCKING
APPLICABLE DOCUMENTS MS9276, MS9581, MS9582

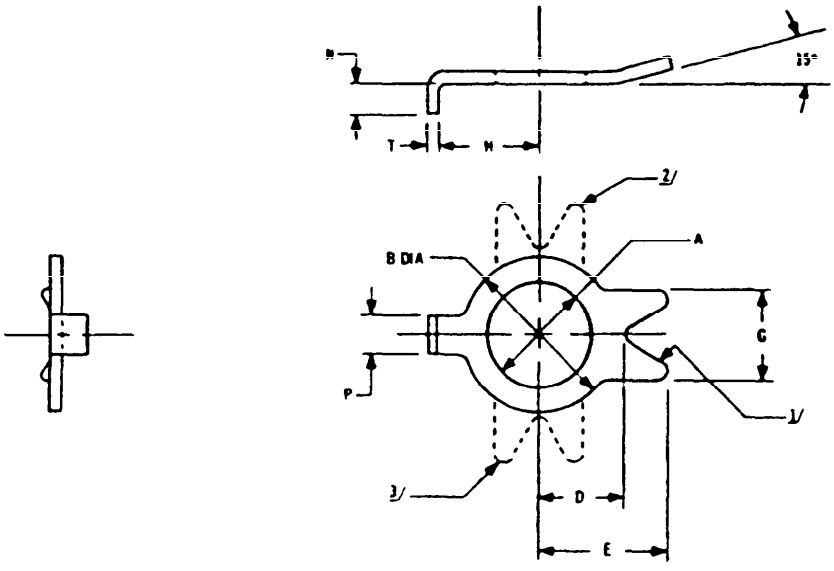


TABLE I. Part numbers.

Material									CRES		
Protective finish									Not specified	Not specified	Not specified
Configuration									1/	2/	3/
Angle									180°	90°	270°
A Min	B Max	D Min	E Min	G Max	H Max	N Min	P Max	T Max	MS9276 - Dash no.	MS9581 - Dash no.	MS9582 - Dash no.
.170	.320	.160	.275	.185	.191	.080	.085	.025	-08	-08	-04
.195	.380	.190	.295	.210	.231	.095	.115	.025	-09	-09	-09
.255	.450	.220	.350	.260	.275	.100	.115	.036	-10	-10	-10
.320	.505	.250	.410	.290	.335	.100	.145	.036	-11	-11	-11
.390	.570	.280	.450	.330	.375	.140	.145	.036	-12	-12	-12
.450	.695	.345	.525	.400	.425	.150	.175	.040	-13	-13	-13
.515	.755	.375	.590	.435	.475	.150	.205	.040	-14	-14	-14
.640	.940	.470	.680	.540	.575	.150	.205	.040	-16	-16	-16
.765	1.065	.530	.730	.560	.675	.150	.235	.040	-17	-17	-17
.890	1.255	.625	.860	.610	.800	.150	.265	.040	-18	-18	-18
1.015	1.440	.715	.950	.660	.875	.150	.295	.040	-19	-19	-19

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 407
WASHER, KEY, LOCKING, SPECIAL DESIGN
APPLICABLE DOCUMENT: MS15820

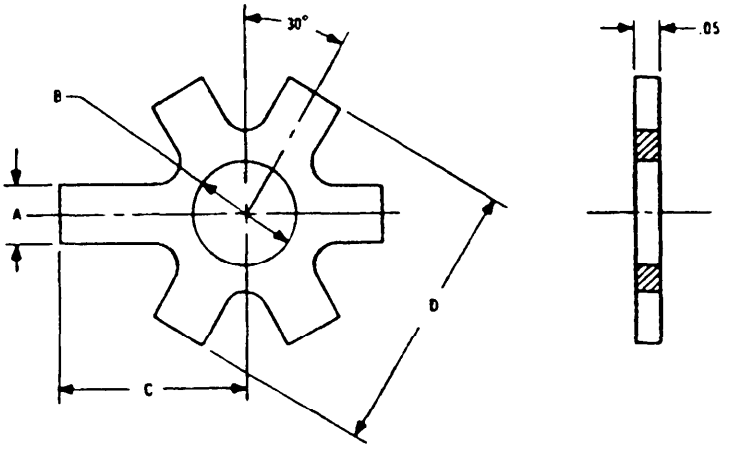


TABLE I. Part numbers.

Material				Steel
Protective finish				Galvanized
B	A	C	D	MS15820 - Dash no.
13/32	1/4	1	1-1/4	-1
17/32	5/16	1-1/8	1-1/2	-2
21/32	3/8	1-1/4	1-3/4	-3
25/32	3/8	1-3/8	2	-4
29/32	5/16	1-1/2	2-1/4	-5
1-1/16	1/2	1-3/4	2-1/2	-6
1-3/16	1/2	1-3/4	2-1/2	-7
1-5/16	1/2	1-7/8	2-5/8	-8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 408
WASHER, KEY, RETAINING, WHEEL BEARING
APPLICABLE DOCUMENT MS21258

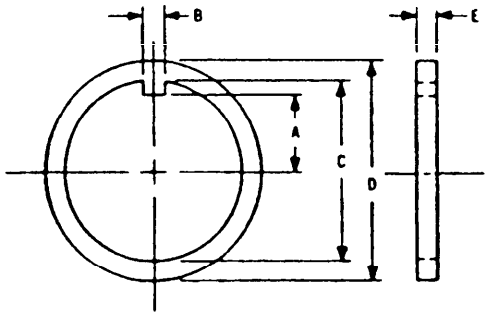


TABLE I. Part numbers.

Material					Steel	
Protective finish					Cadmium plate	Zinc plate
A Min	B Max	C Min	D Max	E Max	MS21258 - Dash no.	
.388	.135	.944	1.405	.098	C15	Z15
.419	.166	1.005	1.530	.130	C16	Z16
.541	.166	1.255	1.780	.130	C20	Z20
.630	.166	1.440	1.905	.130	C23	Z23
.665	.166	1.510	1.905	.130	C24	Z24
.750	.166	1.755	2.280	.130	C28	Z28
.845	.229	1.945	2.405	.130	C31	Z31
.875	.229	2.005	2.655	.130	C32	Z32
1.100	.229	2.445	3.092	.130	C39	Z39
1.302	.291	2.943	3.718	.130	C47	Z47
1.552	.291	3.443	4.780	.130	C55	Z55
2.052	.291	4.445	5.405	.192	C71	Z71

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 409
WASHER, KEY, TAB
APPLICABLE DOCUMENTS MS25081, NAS1636

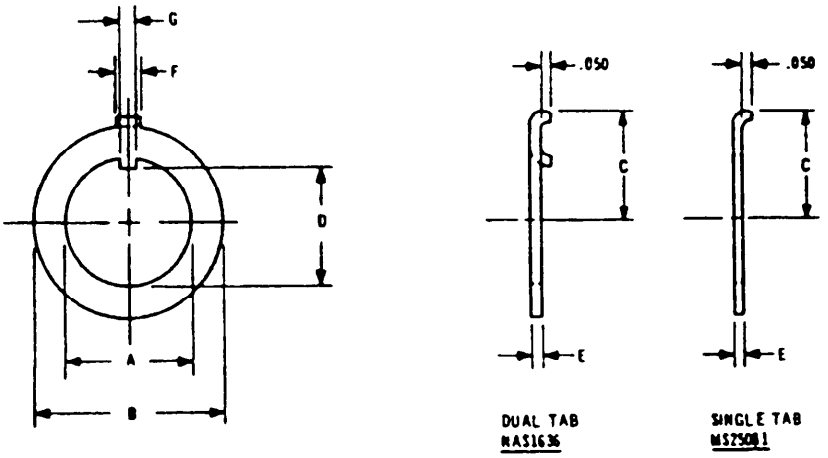


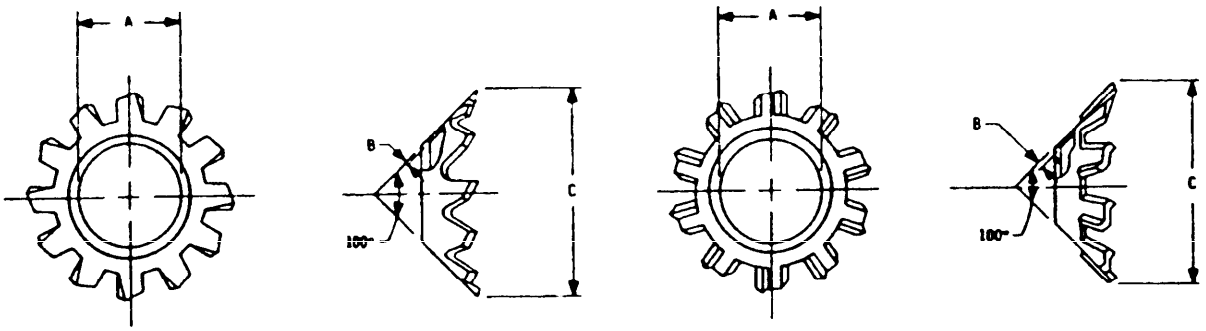
TABLE I. Part numbers.

Material							Carbon steel	CRES	
Protective finish							Cadmium plate	Passivate	
A Min	B Max	C Max	D Min	E Max	F Max	G Max	MS25081 - Dash no.		NAS1636 - Dash no.
.250	.414	.268	.220	.035	.067	.051	-0	C0	—
.377	.642	.370	.340	.035	.115	.065	—	—	-3
.470	.735	.397	.433	.045	.115	.067	-4	C4	—
.470	.735	.417	.433	.035	.115	.065	—	—	-1
.626	.891	.522	.589	.045	.115	.067	-5	C5	—
.626	.891	.494	.589	.035	.115	.065	—	—	-2
.647	.891	.475	.581	.045	.115	.067	-6	C6	—
.747	1.000	.533	.693	.045	.115	.067	-7	C7	—
1.005	1.266	.710	.951	.045	.115	.067	-8	C8	—

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 501
WASHER, LOCK, EXTERNAL TOOTH, COUNTERSUNK
APPLICABLE DOCUMENT MS35790



OPTIONAL DESIGNS

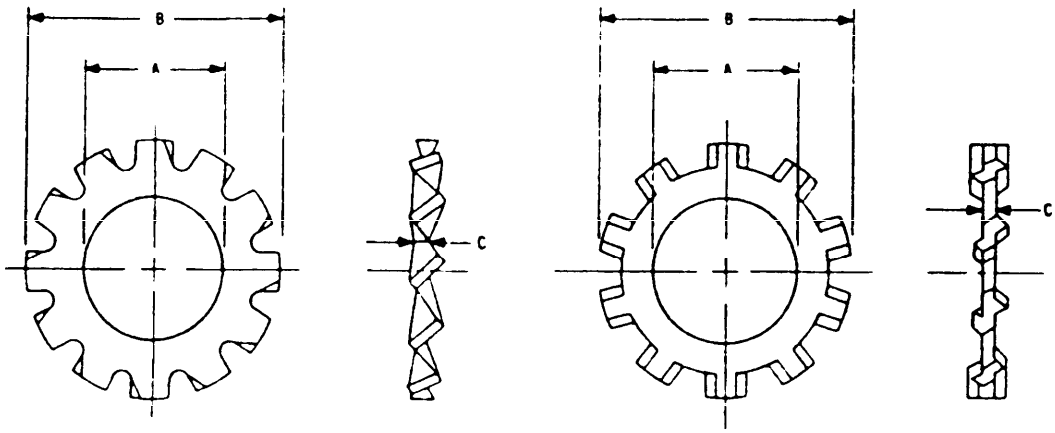
TABLE I. Part numbers.

Material			Carbon steel	Tin brass or phosphor bronze
Protective finish			Cadmium plate	None
A Min	B Max	C Max	MS35790 + Dash no.	
.089	.010	.193	-101	-117
.113	.019	.280	-109	-119
.140	.021	.319	-1	-121
.167	.021	.360	-9	-123
.195	.025	.394	-17	-125
.255	.025	.493	-25	-127
.318	.028	.653	-33	-129
.383	.034	.850	-41	-131

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 502
WASHER LOCK, EXTERNAL TOOTH, FLAT
APPLICABLE DOCUMENT MS35335



OPTIONAL DESIGNS

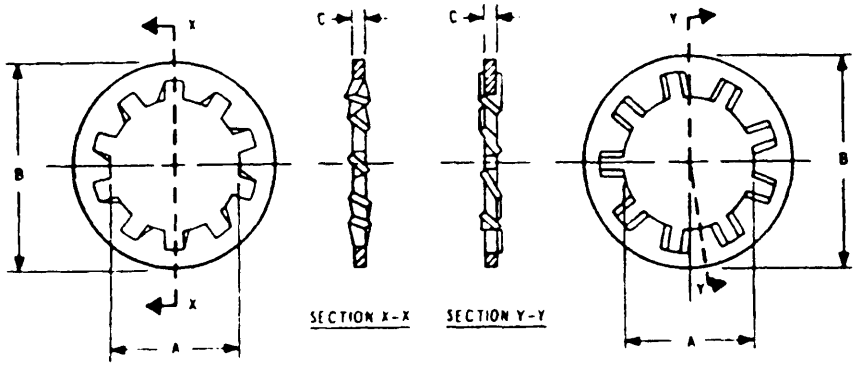
* TABLE I Part numbers.

Material			Carbon steel	CRES	Tin, brass or phosphor bronze
Protective finish			Cadmium plate	Passivate	Uncoated
A Min	B Max	C Max	MS35335 - Dash no.		
.115	.260	.019	-29	-57	-85
.141	.320	.022	-30	-58	-86
.168	.381	.023	-31	-59	-87
.195	.410	.025	-32	-60	-88
.256	.510	.028	-33	-61	-89
.320	.610	.034	-34	-62	-90
.384	.694	.040	-35	-63	-91
.448	.760	.040	-36	-64	-92
.513	.900	.045	-37	-65	-93
.576	.985	.045	-38	-66	-94
.641	1.070	.050	-39	-67	-95
.768	1.260	.055	-40	-68	-96
.897	1.410	.060	-41	-69	-97
1.025	1.620	.067	-42	-70	-98

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 503
WASHER, LOCK, INTERNAL TOOTH, FLAT
APPLICABLE DOCUMENTS MS35333, MS35334



OPTIONAL DESIGNS

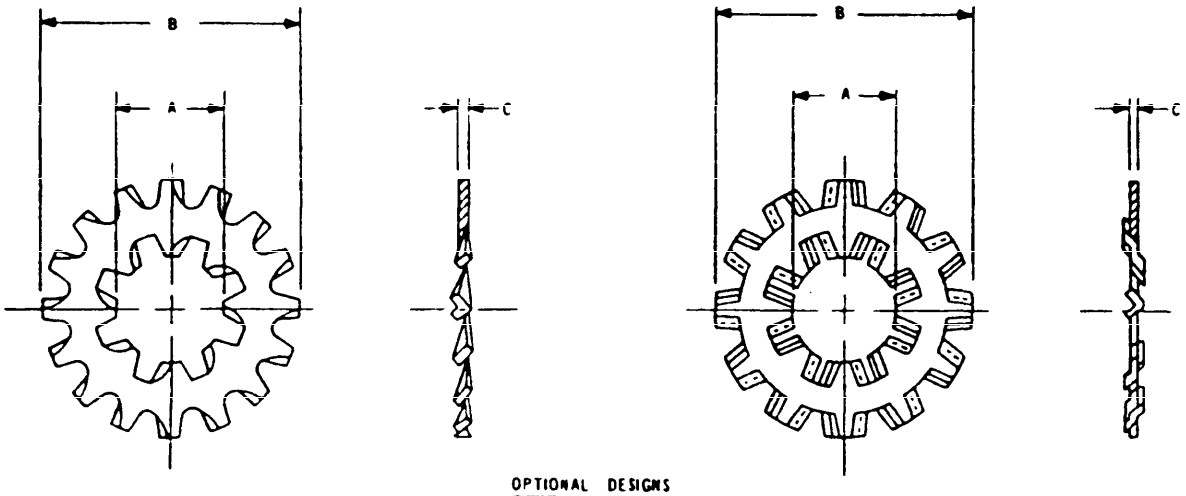
TABLE I Part numbers

Material			Carbon steel		CRES	Tin brass or phosphor bronze
Protective finish			Cadmium plate		Passivate	None
A Min	B Max	C Max	MS35333 - Dash no.	MS35334 - Dash no.	MS35333 - Dash no.	
.089	.200	.015	-35	—	-69	-103
.115	.270	.019	-36	--	-70	-104
.141	.295	.021	-37	—	-71	-105
.168	.340	.023	-38	--	-72	-106
.195	.381	.025	-39	--	-73	-107
.256	.410	.020	-125	--	-135	-145
	.478	.028	-40	—	-74	-108
	.536	.045	--	-19	--	—
.320	.607	.050	—	-20	—	—
	.610	.034	-41	--	-75	-109
.384	.692	.040	-42	--	-76	-110
	.748	.050	--	-21	--	--
.440	.592	.020	-131	--	-141	-151
.448	.789	.040	-43	--	-77	-111
.512	.900	.045	-44	—	-78	-113
	.924	.062	--	-23	--	—
.630	.800	.027	-128	--	-138	-148
.640	1.071	.050	-46	—	-80	-115
	1.135	.062	--	-25	—	--
.768	1.265	.084	--	-26	--	--
.769	1.245	.055	-47	--	-81	-116
.775	1.077	.027	129	--	-139	-149
.894	1.410	.060	-48	--	-82	-117
	1.447	.084	--	-27	--	—
1.010	1.390	.028	-130	--	-140	-150
1.019	1.637	.062	-49	--	-83	-118
1.275	1.975	.062	-51	--	-85	-120

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 504
WASHER LOCK INTERNAL AND EXTERNAL TOOTH, FLAT
APPLICABLE DOCUMENT MS45904



OPTIONAL DESIGNS

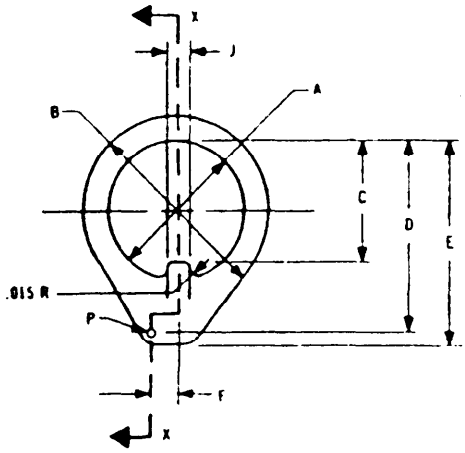
TABLE I. Part numbers.

Material			Carbon steel
Protective finish.			Cadmium plate
A Min	B Max	C Max	MS45904 + Dash no.
115	.475	.021	-51
.141	.510	.028	-54
.168	.610	.034	-57
	.690	.034	-58
	.760	.034	-59
.195	.610	.034	-60
	.690	.040	-61
	.760	.040	-62
	.900	.040	-63
.256	.760	.040	-68
	.900	.040	-69
	.985	.045	-70
	1.070	.045	-71
.320	.900	.040	-72
	.985	.045	-73
	1.070	.050	-74
.384	.985	.045	-76
	1.070	.050	-77
	1.155	.050	-78
	1.260	.050	-79
.512	1.260	.055	-84
	1.410	.060	-86
	1.620	.067	-87
.640	1.410	.060	-92
	1.620	.067	-93

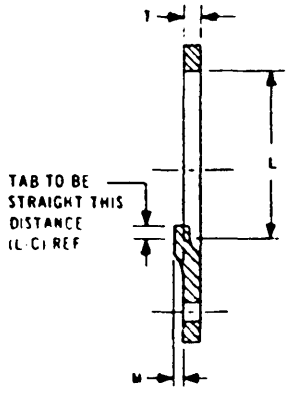
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

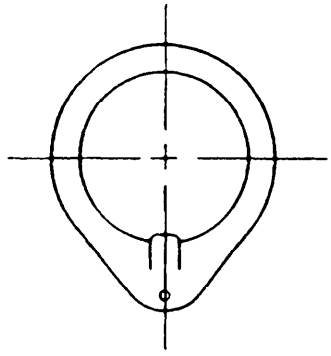
SECTION 505
WASHER, LOCK, ROD END
APPLICABLE DOCUMENT: NAS513



-4 THRU -20



SECTION X-X



-22 THRU -36

TABLE I. Part numbers.

Material											Spring steel
Protective finish											Cadmium plate
Hardness (Rockwell)											C40-45
A	B	C	D	E	F	J	L	M	T	P	NAS513 Dash no.
.272	.44	.214	.406	.500	.125	.052	.293	.036	.050	.062	-4
.334	.50	.273	.484	.578	.125	.052	.355	.036	.050	.062	-5
.396	.56	.327	.562	.656	.125	.082	.418	.036	.050	.062	-6
.459	.63	.366	.625	.719	.125	.082	.463	.049	.063	.062	-7
.520	.75	.450	.719	.813	.156	.082	.542	.049	.063	.062	-8
.647	1.00	.559	.937	1.031	.188	.114	.675	.057	.071	.071	-10
.772	1.12	.681	1.062	1.156	.188	.114	.800	.057	.071	.071	-12
.897	1.31	.795	1.250	1.344	.188	.142	.944	.066	.080	.089	-14
1.022	1.50	.918	1.422	1.516	.188	.142	1.088	.074	.090	.090	-16
1.147	1.62	1.028	1.547	1.656	.219	.174	1.213	.074	.090	.090	-18
1.272	1.75	1.154	1.687	1.796	.219	174	1.280	.096	.112	.112	-20
1.397	1.88	1.254	1.859	1.969	--	.236	1.405	.096	112	.112	-22
1.518	2.00	1.379	1.969	2.078	--	.236	1.530	.096	.112	.112	-24
1.643	2.12	1.495	2.109	2.219	--	.236	1.638	.109	.125	.125	-26
1.766	2.25	1.607	2.250	2.375	--	.298	1.763	.109	.125	.125	-28
1.891	2.38	1.732	2.375	2.484	--	.298	1.888	.109	.125	.125	-30
2.016	2.63	1.857	2.594	2.703	--	.298	2.076	.109	.125	.125	-32
2.141	2.75	1.973	2.750	2.860	--	.298	2.201	.109	.125	.125	-34
2.266	2.88	2.098	2.875	2.984	--	.298	2.326	.109	.125	.125	-36

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 506
WASHER LOCK SPRING HELICAL
APPLICABLE DOCUMENTS MS35338, MS35340, MS51848

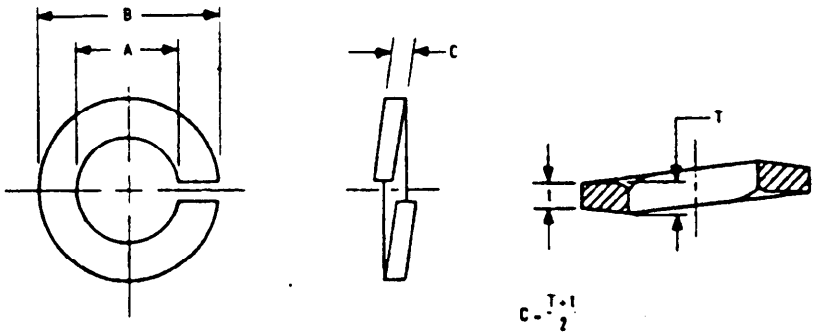


TABLE I. Part numbers.

Material			Carbon steel					CRES Series					Ph brz or tin brs	Nickel copper alloy
								300	300	420	300	420		
Protective finish			Cadmium plate			Phosphate		Passivate			Black oxide		Cadmium plate	None
Hardness (Rockwell)			C45-43	C45-53	C45-53	Not specified		C35-43	C35-43	C45-53	C35-43	C45-53	None	
A Min	B Max	C Min	MS51848 - Dash no.	MS35338 - Dash no.	MS35340 - Dash no.	MS35338 - Dash no.	MS51848 - Dash no.	MS51848 - Dash no.	MS35338 - Dash no.					
.062	.092	.016	-1	—	—	—	-21	-41	—	—	—	—	--	—
.088	.135	.020	-3	—	—	—	-23	-43	—	—	—	—	—	—
	.172	.020	--	-39	—	-58	—	—	-134	-153	-1348	-1538	-96	-115
	.208	.027	—	--	-39	—	—	—	—	—	—	—	—	—
.114	.173	.022	-5	--	—	—	-25	-45	—	—	—	—	—	—
	.115	.034	—	—	-40	—	—	—	—	—	—	—	—	—
.115	.209	.025	--	-40	--	-59	--	—	-135	-154	-1358	-1548	-97	-116
.141	.216	.030	-7	--	—	—	-27	-47	—	—	—	—	--	—
	.250	.031	—	-41	--	-60	—	—	-136	-155	-1368	-1558	-98	-117
	.314	.045	—	—	-41	—	—	—	—	—	—	--	—	—
.167	.267	.047	-8	—	--	—	-28	-48	—	—	—	—	—	—
	.168	.057	--	--	-42	—	—	—	—	—	—	—	--	—
.168	.293	.040	—	-42	—	-61	—	—	-137	-156	-1378	-1568	-99	-118
.193	.294	.047	-9	—	--	—	-29	-49	—	—	--	—	—	—
	.194	.068	--	—	-43	--	—	—	—	--	—	--	--	—
.194	.334	.047	—	-43	—	-62	—	—	-138	-157	-1388	-1578	-100	-119
.254	.365	.078	-10	--	--	--	-30	-50	—	—	—	—	--	--
	.255	.084	—	--	-44	—	—	--	--	—	—	--	--	—
.255	.489	.062	—	-44	--	-63	--	—	-139	-158	-1398	-1588	-101	-120
.317	.460	.093	-11	--	--	--	-31	-51	—	—	--	--	--	--
	.318	.108	--	--	-45	--	--	--	—	--	—	--	--	--

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE I Part numbers. - Continued

Material			Carbon steel					CRES					Phosphor or tin bronze	Nickel copper alloy
								Series						
								300	300	420	300	420		
Protective finish			Cadmium plate			Phosphate		Passivate			Black oxide		Cadmium plate	None
Hardness (Rockwell)			C45-53	C45-53	C45-53	C45-53		C35-43	C35-43	C45-53	C35-43	C45-53	None	
A Min	B Max	C Min	MS51848 - Dash no.	MS35338 - Dash no.	MS35340 - Dash no.	MS35338 - Dash no.	MS51848 - Dash no.	MS51848 - Dash no.	MS35338 - Dash no.					
.318	.586	.078	--	-45	—	-64	—	--	-140	-159	-140B	-159B	-102	-121
.382	.553 .741	.125 .123	-12 —	— --	-- -46	— --	-32 --	-52 —	-- --	-- --	-- --	— --	-- --	-- --
.382	.683	.094	--	-46		-65	--	--	-141	-160	-141B	-160B	-103	-122
.443	.647	.140	-13	--	--	--	-33	-53	—	--	--	--	--	--
.446	.779	.109	--	-47	--	-66	--	—	-142	-161	-142B	-161B	-104	-123
.509	.737 .939	.172 .162	-14 —	— --	— -48	— --	-34 --	-54 —	— —	— —	— --	— --	— --	— --
.509	.873	.125	--	-48	—	-67	—	—	-143	-162	-143B	-162B	-105	-124
.572	.971	.141	--	-49	—	-68	—	—	-144	-163	-144B	-163B	-106	-125
.636	.923 1.157	.203 .202	-15 —	— —	— -50	— —	— —	-55 —	— —	— —	— —	— —	— —	— —
.636	1.079	.156	--	-50	—	-69	—	—	-145	-164	-145B	-164B	-107	-126
.763	1.111 1.361	.218 .241	-16 --	— --	— -51	— —	— —	— —	— —	— —	— --	— --	— —	— —
.763	1.271	.188	--	-51	—	-70	—	—	-146	-165	-146B	-165B	-108	-127
.887	1.296	.234	-17	--	—	—	—	--	—	--	--	--	--	--
.890	1.464	.219	—	-52	--	-71	—	—	-147	-166	-147B	-166B	-109	-128
1.017	1.483 1.661 1.799	.250 .250 .330	-18 -- —	— -53 —	— -- -53	— -72 —	— — —	— — —	— -148 --	— -167 --	— -148B —	— -167B --	— -110 —	— -129 --
1.144	1.853	.281	—	-54	—	--	—	--	-149	-168	-149B	-168B	-111	-130
1.271	2.045 2.231	.312 .417	— --	-55 --	-- -55	— —	— —	— —	-150 —	-169 —	-150B --	-169B —	-112 —	-131 —
1.398	2.239	.344	--	-56	—	—	--	—	-151	-170	-151B	-170B	-113	-132
1.525	2.430 2.638	.375 .496	-- --	-57 —	-- -57	-- --	— --	— —	-152 —	-171 —	-152B —	-171B —	-114 —	-133 —

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T07:52Z
Check the source to verify that this is the currenversion before use.

SECTION 601
WASHER, RECESSED, COUNTERSUNK, ONE SURFACE
APPLICABLE DOCUMENTS: NAS1587, MS20002, MS21206

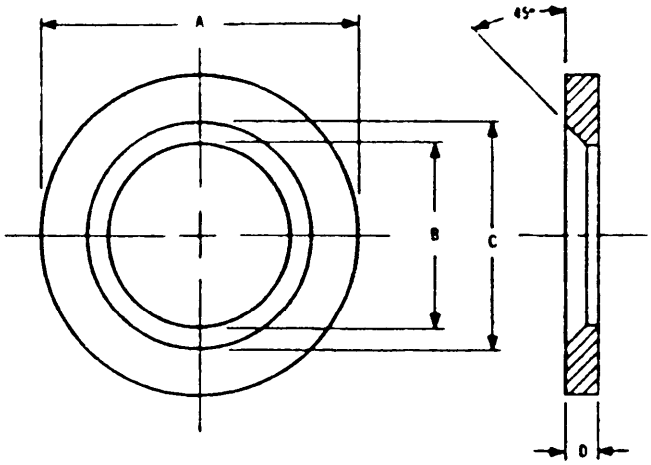


TABLE I. Part numbers.

Material				CRES Type 321 or 347	Alloy steel	
Protective finish				Passivate	Cadmium plate	
Tensile strength (psi)				75,000 Min	125,000 to 145,000	Not specified
Hardness (Rockwell)				Not specified	Not specified	C38-45
B Min	C Min	A Max	D Max	NAS1587 + Dash no.	MS20002 + Dash no.	MS21206 + Dash no.
.191	.280	.372	.069	—	—	C3
.192	.222	.479	.072	-3C	—	—
.251	.340	.470	.069	—	—	C4
.252	.334	.541	.088	-4C	C4	—
.314	.402	.573	.069	—	—	C5
.315	.356	.593	.078	-5C	C5	—
.376	.497	.679	.085	—	—	C6
.378	.483	.697	.088	-6C	C6	—
.439	.559	.783	.085	--	—	C7
.441	.543	.791	.088	-7C	C7	—
.501	.622	.898	.085	—	—	C8
.504	.604	.085	.088	-8C	C8	—
.627	.778	1.125	.085	—	—	C10
.631	.765	1.072	.088	-10C	C10	—
.752	.903	1.350	.085	—	—	C12
.757	.890	1.260	.088	-12C	C12	—
.877	1.028	1.572	.085	--	—	C14
.884	1.015	1.447	.088	-14C	C14	—
1.002	1.153	1.798	.119	--	--	C16
1.010	1.140	1.635	.088	-16C	C16	—
1.127	1.278	2.188	.119	--	—	C18
1.135	1.265	1.885	.088	-18C	C18	—
1.252	1.435	2.260	.119	—	—	C20
1.260	1.427	2.135	.104	-20C	C20	—
1.377	1.560	2.490	.119	--	—	C22
1.385	1.552	2.323	.104	--	C22	—
1.502	1.685	2.723	.119	—	—	C24
1.510	1.677	2.510	.104	—	C24	—

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 602
WASHER, RECESSED, COUNTERSUNK, TWO SURFACES
APPLICABLE DOCUMENTS: MS9482, MS9768, MS14155, MS14177

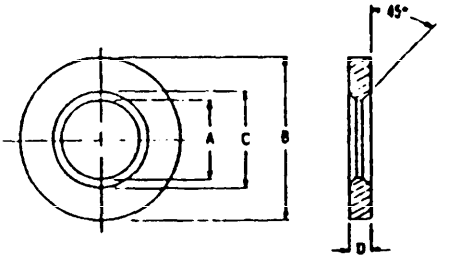


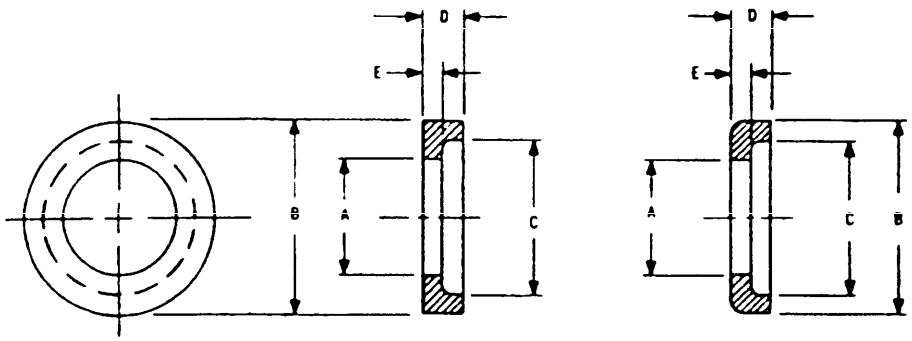
TABLE I. Part numbers.

Material				Steel	CRES	Alloy steel	
Protective finish				Diffused nickel-cadmium plate	None	Cadmium plate	
Hardness (Rockwell)				C50-56	C25-39	C39-43	
A Min	B Max	C Min	D Max	MS9482 • Dash no.	MS9768 • Dash no.	MS14155 • Dash no.	MS14177 • Dash no.
.140	.340	.176	.076	—	-07	—	—
.143	.315	.178	.070	-07	—	—	—
.166	.385	.202	.076	—	-08	—	—
.169	.375	.214	.085	-08	—	—	—
.192	.448	.236	.085	—	-09	—	—
.195	.435	.250	.095	-09	—	—	—
.251	.505	.337	.124	—	—	-4	—
	.532	.337	.133	—	—	—	-4
.252	.635	.296	.085	—	-10	—	—
.256	.570	.332	.135	-10	—	—	—
.314	.620	.399	.124	—	—	-5	—
	.651	.399	.133	—	—	—	-5
.315	.760	.359	.105	—	-11	—	—
.317	.700	.394	.160	-11	—	—	—
.376	.766	.494	.158	—	—	-6	—
	.795	.494	.167	—	—	—	-6
.378	.885	.422	.117	—	-12	—	—
.380	.855	.489	.195	-12	—	—	—
.439	.878	.556	.158	—	—	-7	—
	.912	.556	.167	--	—	—	-7
.441	1.010	.495	.130	—	-13	—	—
.443	.980	.552	.215	-13	—	—	—
.501	1.004	.619	.163	--	—	-8	—
	1.038	.619	.167	—	—	--	-8
.504	1.198	.558	.165	—	-14	—	—
.505	1.170	.614	.260	-14	—	—	—
.627	1.266	.775	.226	—	—	-10	—
	1.302	.775	.198	--	--	—	-10
.630	1.405	.771	.315	-16	—	—	—
.752	1.507	.900	.278	--	—	-12	—
	1.546	.900	.198	--	—	—	-12
.755	1.670	.896	.375	-17	--	—	—
.877	1.747	1.025	.320	—	—	-14	—
	1.791	1.025	.198	—	--	—	-14
.880	1.935	1.021	.430	-18	—	—	—
1.002	1.978	1.150	.368	—	—	-16	—
1.005	2.195	1.146	.490	-19	—	—	—

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 603
WASHER, RECESSED, TAPER PIN
APPLICABLE DOCUMENT AN975



OPTIONAL DESIGN

TABLE I. Part numbers

Material					Steel
Protective finish					Cadmium plate
A	B	C	D	E	AN975 - Dash no.
13/64	15/32	11/32	3/16	1/16	-3
17/64	9/16	13/32	13/64	5/64	-4
21/64	11/16	17/32	13/64	5/64	-5
25/64	25/32	19/32	13/64	5/64	-6
29/64	27/32	21/32	13/64	5/64	-7
33/64	15/16	23/32	7/32	3/32	-8
49/64	1 9/32	1	15/64	7/64	-12
57/64	1 15/32	1 3/16	15/64	7/64	-14

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 701
WASHER SHOULDERED AND RECESSED DIMPLED 100
APPLICABLE DOCUMENT NAS1169

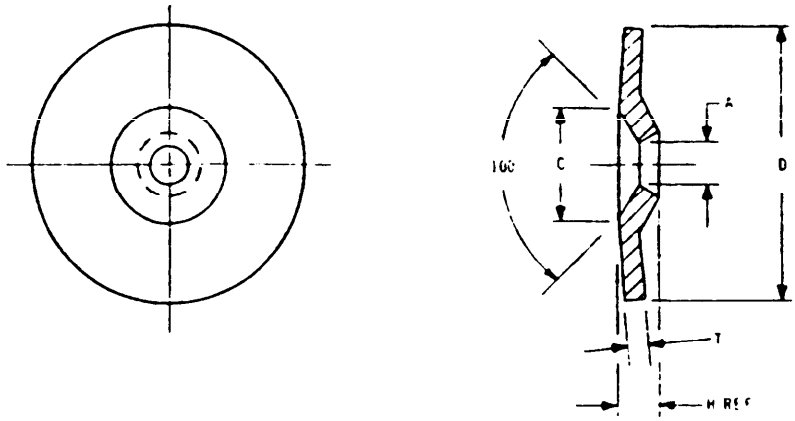


TABLE I Part numbers

Material	Carbon steel	Aluminum alloy		CRES	Titanium
		5052	2024		
Protective finish	Cadmium plate	Anodize		Passivate	Not specified
Hardness or tensile strength (ksi)	Rockwell C42-47	Not specified		160 tensile	None

A Min	D Max	C Min	H Ref	T	NAS1169 - Dash no.				
.140	.495	.279	.068 .076 .082	.008 .016 .025	-- -6L -6	D6E -- --	-- DD6L DD6	-- C6L C6	-- T6L T6
.166	.582	.332	.079 .087 .094	.008 .016 .025	-- -8L -8	D8E -- --	-- DD8L DD8	-- C8L C8	-- T8L T8
192	.645	.385	.091 .099 .101	.008 .016 .025	-- -10L -10	D10E -- --	-- DD10L DD10	-- C10L C10	-- T10L T10
.252	.770	.507	.117 .125 .132	.008 .016 .025	-- -416L -416	D416E -- -	-- DD416L DD416	-- C416L C416	-- T416L T416
.314	.958	.635	.145 .153 .160	.006 .016 .025	-- -516L -516	D516E -- --	-- DD516L DD516	-- C516L C516	-- T516L T516
.377	1.020	.762	.172 .180 .188	.008 .016 .025	-- -616L -616	D616E -- --	-- DD616L DD616	-- C616L C616	-- T616L T616
440	1.208	.890	.200 .208 .215	.008 .016 .025	-- -716L -716	D716E -- --	-- DD716L DD716	-- C716L C716	-- T716L T716
502	1332	1.017	.225 .237 .245	.008 .016 .025	-- -816L -816	D816E -- -	-- DD816L DD816	-- C816L C816	-- T816L T816

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 801
WASHER, SPRING TENSION
APPLICABLE DOCUMENT: MIL-W-12133/1

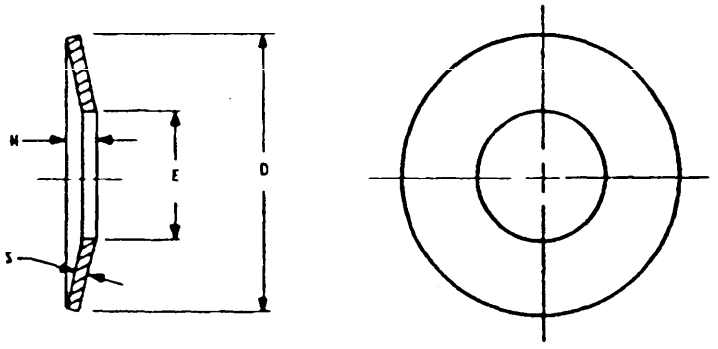


TABLE I. Part numbers.

Material				Spring steel	
Protective finish				Plain	Cadmium plate
E Min	D Max	S Nom	H Nom	MIL-W-12133/1 Dash no.	
.125	.250	.013	.020	-1	-1P
.156	.312	.017	.025	-2	-2P
.195	.375	.015	.027	-3	-3P
.250	.500	.020	.035	-4	-4P
.317	.625	.032	.051	-5	-5P
.382	.750	.032	.054	-6	-6P
.445	1.000	.035	.067	-7	-7P
.512	1.000	.039	.075	-8	-8P
	1.102	.048	.083	-9	-9P
.637	1.375	.059	.102	-10	-10P
.761	1.500	.059	.114	-11	-11P
		.078	.121	-12	-12P
.880	1.750	.085	.128	-13	-13P
1.016	2.000	.079	.138	-14	-14P
	2.375	.098	.177	-15	-15P

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

...tems Division (AFSC)
...n AFB, OH 45433

POSTAGE AND FEES PAID

U.S. MAIL

...SINESS
...TE USE $300

Commander
Aeronautical Systems Division (AFSC)
ATTN: ASD/ENESS
Wright-Patterson AFB, OH 45433

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-STD-1764 WASHERS, PREFERRED FOR DESIGN, LISTING OF

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address – Optional)*

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:52Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
Washington, DC 20301

Washers Preferred For Design, Listing of

MIL-STD-1764

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Aeronautical Systems Division (AFSC), ATTN: ASD/ENESS, Wright-Patterson Air Force Base, Ohio 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

FOREWORD

1. The purpose of this bookform standard is to provide a commodity type parts document on washers to aid military equipment designers and engineers in the selection of preferred washers.

2. This document consists of an index of preferred standardization documents and a listing of preferred parts within these documents that have been selected with respect to configuration, sizes, lengths, materials, and finishes for washers.

3. The selection of preferred documents listed in this standard and the selection of part numbers within the preferred documents were made as follows:

a. Selection of Documents

(1) Documents listed or scheduled for listing in the Department of Defense Index of Specifications and Standards (DODISS).

(2) Documents which are active for design.

(3) Documents specifying part numbers (dash numbers) which designate specific sizes, materials and finishes.

b. Selection of Part Numbers

(1) By conducting a thorough search and evaluation of existing DoD procurement information.

(2) By evaluation of preferred parts listed in recent weapon system contracts.

(3) By evaluation of preferred parts lists obtained from industry.

4. To increase the scope and versatility of this washer standard, periodic revisions will be developed. Results from Standardization studies, MILITARY PARTS CONTROL ADVISORY GROUP (MPCAG) evaluations, evaluation of a new family of washers and recommendations from interested activities will form the basis for these revisions.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:52Z
Check the source to verify that this is the current version before use.

CONTENTS

			Page
Paragraphs	1.	SCOPE	1
	2.	REFERENCED DOCUMENTS	1-3
	3.	DEFINITIONS	3-4
	4.	GENERAL STATEMENTS	4
	5.	DETAILED REQUIREMENTS	4
	6.	NOTES	5

SECTIONS

Section	100	WASHER, BEVEL	
	101	Washer, Bevel, 9-1/2°	101.1
	200	WASHER, FINISHING	
	201	Washer, Finishing, Countersunk	201.1
	202	Washer, Finishing, Flush	202.1
	300	WASHER, FLAT	
	301	Washer, Flat, Rectangular	301.1
	302	Washer, Flat, Round	302.1
	303	Washer, Flat, Square	303.1
	304	Washer, Flat, Square with Radius	304.1
	400	WASHER, KEY	
	401	Washer, Key, Bearing	401.1
	402	Washer, Key, Cup, Lock	402.1
	403	Washer, Key, Cup, Lock, Spanner Nut	403.1
	404	Washer, Key, Drive (Synchro)	404.1
	405	Washer, Key, Flat, General Purpose	405.1
	406	Washer, Key, Locking	406.1
	407	Washer, Key, Locking, Special Design	407.1
	408	Washer, Key, Retaining, Wheel Bearing	408.1
	409	Washer, Key, Tab	409.1
	500	WASHER, LOCK	
	501	Washer, Lock, External Tooth, Countersunk	501.1
	502	Washer, Lock, External Tooth, Flat	502.1
	503	Washer, Lock, Internal Tooth, Flat	503.1
	504	Washer, Lock, Internal and External Tooth, Flat	504.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTIONS - Continued

			Page
Section	505	Washer, Lock, Rod End	505.1
	506	Washer, Lock, Spring, Helical	506.1
	600	WASHER, RECESSED	
	601	Washer, Recessed, Countersunk, One Surface	601.1
	602	Washer, Recessed, Countersunk, Two Surfaces	602.1
	603	Washer, Recessed, Taper Pin	603.1
	700	WASHER, SHOULDERED AND RECESSED	
	701	Washer, Shouldered and Recessed, Dimpled, 100°	701.1
	800	WASHER, SPRING TENSION	
	801	Washer, Spring Tension	801.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard provides a listing of preferred washers encompassing the following characteristics:

a. Configuration

b. Size

c. Materials

d. Protective Coatings and Finishes

1.2 Purpose. The purpose of this standard is as follows:

a. Provide the designer with a listing of preferred washers to promote their use in design of weapon systems and equipments.

b. Control and minimize the variety of washers used in military equipment thereby facilitating logistic support of the equipment during its life cycle.

1.3 Application. To minimize the proliferation of washers, only the preferred part number listed herein is authorized for use in new design. All other part numbers, even though shown on current Military Specification Sheets, Military Standards (MS), National Aerospace Standards (NAS), Aeronautical Standards (AS), and Air Force/Navy Aeronautical Standards (AN), are not approved for use in new design unless approved by the cognizant Government procuring activity.

1.4 Intended Use. Implement this standard by including one of the following options in the contract:

a. Require this standard as a supplement to an end use type standard such as MIL-STD-1471 or MIL-STD-1515. When thus required, only the washers listed in both the end use type and this standard are acceptable. Use of other washers requires approval of the Government procuring activity.

b. Require this standard as a guide to be used with an end use type standard such as MIL-STD-1471 or MIL-STD-1515. When thus required, the washers listed in the end use type standard and this standard are acceptable. The designer must assure himself the washers listed in both the end use type standard and this standard are not adequate for his requirement before using washers not listed herein. Use of washers not listed in the end use type standard requires approval of the Government procuring activity.

c. Require this standard and indicate exceptions to it. When thus required, only the washers listed in this standard and not excluded by the exceptions are acceptable. Use of other washers requires approval of the Government procuring activity.

d. Require this standard as a guide. When thus required, the designer must assure himself the washers listed in this standard are not adequate for the requirement before using other washers.

2. REFERENCED DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-12133/1 - Washer, Spring Tension (Belleville Spring).

STANDARDS

MILITARY

MS9276 - Washer, Key-Cres, AMS 5510, 180° Locking.

MS9320 - Washer, Flat - AMS 6350.

MS9321 - Washer, Flat - AMS 5510.

MS9482 - Washer, Flat - Steel AMS 6437 or AMS 6485, Diffused Nickel - Cadmium Plated, Countersunk.

MS9549 - Washer, Flat - AMS 5510.

MS9581 - Washer, Key-Cres, AMS 5510, 90° Locking.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY - Continued

MS9582 - Washer, Key-Cres, AMS 5510, 270° Locking.

MS9768 - Washer, Flat-Cres, AMS 5525 or AMS 5737, Countersunk.

MS9880 - Washer, Cup, Lock-Cres, AMS 5510.

MS9952 - Washer, Cup, Lock-Spanner Nut Cres, AMS 5646.

MS14151 - Washer, Flat, Round, Corrosion Resisting Steel (for Electrical Applications).

MS14155 - Washer, Recessed Countersunk Two Surfaces, for Use with Wheel Bolts and Nuts up to and Including 180 KSI Ftu.

MS14177 - Washer, Recessed, Countersunk Two Surfaces, for Use with Wheel Bolts and Nuts up to and Including 220 KSI Ftu.

MS15795 - Washer, Flat-Metal, Round, General Purpose (IN./MM).

MS15820 - Washers, Lock Special Design, Galvanized Steel.

MS16212 - Washer, Flat, Metal, Round, Medium Series, Nonmagnetic.

MS17186 - Washer, Drive (Synchro).

MS19069 - Washers, Key, Retaining, Ball and Roller Bearing, Light Series.

MS19070 - Washers, Key, Retaining, Ball and Roller Bearing, Regular Series.

MS20002 - Washer, Countersunk and Plain, High Strength.

MS21206 - Washer, Countersunk and Plain, for Use with Bolts and Nuts up to and Including 220 KSI Ftu.

MS21258 - Washer, Key, Retaining, Wheel Bearing.

MS21307 - Washer, Flat (Square).

MS25081 - Washer, Key.

MS25440 - Washers, for Use with Aircraft Aluminum Terminals.

MS27111 - Washer, Key-Flat, General Purpose.

MS27129 - Washer, Finishing-Countersunk.

MS27183 - Washer, Flat-Round, Steel, Cadmium Plated, General Purpose (IN./MM).

MS35333 - Washer, Lock, Flat-Internal Tooth.

MS35334 - Washer, Lock-Flat, Heavy, Internal Tooth.

MS35335 - Washer, Lock, Flat-External Tooth (IN./MM).

MS35338 - Washer, Lock-Spring, Helical, Regular (Medium) Series, (IN./MM).

MS35340 - Washer, Lock-Helical Spring, Extra Duty.

MS35790 - Washer, Lock-Countersunk, 100°, External Tooth (IN./MM).

MS45904 - Washer, Lock, Internal and External Tooth.

MS51848 - Washer, Lock-Helical Spring, Hi-Collar.

MS51859 - Washer, Flat-Plastic (Nylon).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY - Continued

MS51867 - Washer, Metal, Flat, Plain, Rectangular.

MS172271 thru
MS172320 - Washer, Key-Single, Bearing Retaining, AMS 6350.

AIR FORCE-NAVY AERONAUTICAL

AN960 - Washer, Flat.

AN961 - Washer-Flat, Electrical.

AN970 - Washer, Flat, for Wood.

AN975 - Washer-Taper Pin.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC. (AIA)

NATIONAL AEROSPACE STANDARDS

NAS390 - Washer-Flush Type Finishing.

NAS513 - Washer-Rod End Locking.

NAS549 - Washer, Nonmetallic-Insulating, Electrical.

NAS620 - Washer, Flat-Reduced Outside Diameter.

NAS1099 - Washer-Bevel 9-1/2°.

NAS1169 - Washer, Dimpled, 100°.

NAS1197 - Washer - 5052 Aluminum Alloy.

NAS1252 - Washer - 7075 Aluminum Alloy Flat.

NAS1401 - Washer, Radius.

NAS1515 - Washers-Plastic and Synthetic Rubber.

NAS1587 - Washer Plain and CSK, 1200°F.

NAS1636 - Washer, Key, Dual Tab.

(Application for copies should be addressed to the Aerospace Industries Association of America, Inc., 1725 De Sales Street, N.W., Washington, DC 20036.)

3. DEFINITIONS

3.1 Adopted Industry Standards. Any Industry Specification or Standard which is listed in the Department of Defense Index of Specifications and Standards (DODISS).

3.2 Commodity Type Document. A document which lists preferred parts within a Federal Supply Classification class or Item Name. This document is to be used for selecting preferred parts for a new design when the document is invoked as a contractual requirement in conjunction with a parts control requirement.

3.3 End Use Type Document. A document that lists preferred documents and establishes parts requirements which are contractually binding for the design and construction/manufacture of a weapon system or an established equipment category such as MIL-STD-1515.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:52Z
Check the source to verify that this is the currentversion before use.

3.4 Military Parts Control Advisory Group (MPCAG). A Department of Defense organization which provides advice to the Military Departments and military contractors on the selection of parts in assigned commodity classes, and collects data on nonstandard parts for developing or updating military specifications and standards.

3.5 Definitions of approved item names used in this standard are as follows:

a. Washer. A perforated piece of metal, leather, or other material of various shapes and forms. It is used primarily as a seat for a bolt, nut, screw, rivet, or nail to distribute pressure and to prevent endwise motion over an area surrounding a hole; to increase wear resistance in that area, and/or to prevent movement between two parts by means of restraining features created by the shape of its edges.

(1) Washer, Bevel. An item having a centrally located hole and two flat bearing surfaces which are not parallel.

(2) Washer, Finishing. An item that is flush flanged and of a cone or disk shape with a centrally located hole.

(3) Washer, Flat. An item with a centrally located hole, a solid (not laminated) cross-section and two flat parallel bearing surfaces. The flat parallel bearing surfaces may have a rough or machine finish, or a specified surface designation between 32 and 125 microinch arithmetical average (AA) if required for specified high torque wrenching requirements, but must not have any of the following conditions: visible or detectable means of lubrication; grooves; indentations or self-lubrication. The dimension of the centrally located hole must be 75 percent of the outside diameter or less. The thickness must be 0.006 inch or more for items less than 1/4 inch outside diameter and 0.016 inch or more for items with 1/4 inch or more outside diameter. The thickness shall not exceed 25 percent of the outside diameter or the shortest distance between peripheral flats. The maximum limit of the tolerance or range applied to the thickness and/or outside diameter must be used to determine compliance. Do not use nominal values.

(4) Washer, Key. A washer of nonresilient material usually having two parallel bearing surfaces. The outside and/or inside edges have one or more protuberances. The ends and/or edges may be prebent at any angle to the bearing surfaces. The keys or lugs may be fitted into keyways in adjacent material or they may be bent over after application to hold or lock one or more items in place.

(5) Washer, Lock. An item which is usually round with a centrally located hole. Its edges and two bearing surfaces are so designed that applied pressure brings into play the spring tension and/or frictional locking principle which materially resists any tendency toward turning.

(6) Washer, Recessed. An item round in shape and having a centrally located hole. Both bearing surfaces are parallel and flat, but one or both of the bearing surfaces must be broken by a counterbore, countersink, or similarly shaped recess.

(7) Washer, Shouldered and Recessed. An item round in shape having a centrally located hole. One bearing surface is flat, the other surface is in the form of a raised boss (shoulder) having a flat bearing surface. Either or both of the bearing surfaces must be broken by a counterbore, countersink or similar shaped recess. The thickness of the boss must not exceed 25 percent of the boss diameter.

(8) Washer, Spring Tension. A bent spring metal washer with a centrally located hole. It is designed to exert pressure axially when assembled in place.

Note: Reference must be made to 3.5.a above (the indicated name), in order to understand the basic concept which is inherent in the total concept of the item names of 3.5.a.(1) thru 3.5.a.(8) above.

4. GENERAL STATEMENTS

4.1 Selection Procedure

4.1.1 Document Selection. The applicable section shall be selected after reviewing the table of contents.

4.1.2 Part Number Selection (Preliminary). A preliminary selection of the applicable part number shall be made after reviewing the nominal parameters (sizes, materials, tensile strength) listed in the sections.

4.1.3 Part Number Selection (Final). A final selection of the applicable part number shall be made after reviewing the detailed requirements specified in the referenced washer documents for suitability in the particular military equipment being designed (considering the application and environmental conditions).

5. DETAILED REQUIREMENTS

5.1 The detailed requirements for preferred washers are contained in the applicable washer document and associated procurement specification. If there is disagreement between the nominal parameters listed in this standard and the parameters specified in the applicable washer document or associated procurement specification, the parameters specified in the applicable washer document or associated procurement specification shall prevail.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Dimensions. Dimensions shown in the sections contained herein are in inches.

6.1.1 Unreferenced dimensions are nominal.

Custodians:
Army - AR
Navy - None
Air Force - 11

Review Activities:
Army - EA, MI
Navy - OS
Air Force - None
DLA - IS
NS

User Activities
Army - ME
Navy - EC, SH
Air Force - None

Preparing activity:
Air Force - 11

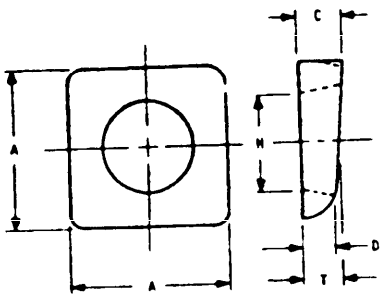
Agent:
DLA - IS

(Project 5310-0904)

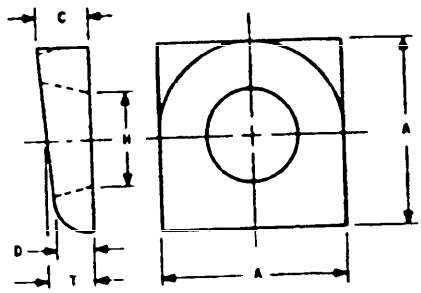
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 101
WASHER BEVEL, 9-1/2°
APPLICABLE DOCUMENT NAS1099



PARTS -4, -5, -6, -8



PARTS 10, -12, -14, -16, -18

TABLE I. Part numbers.

Material					Malleable iron - cast
Protective finish					Cadmium plate
H Min	A	T	C	D	NAS1099 + Dash no
.28	.75	.19	.25	.13	-4
.35	1.00	.19	.27	.11	-5
.47	1.25	.28	.34	.19	-6
.63	1.31	.31	.38	.22	-8
.72	1.56	.28	.36	.13	-10
.81	1.56	.28	.41	.16	-12
.94	1.75	.28	.44	.19	-14
1.06	1.81	.28	.44	.19	-16
1.22	2.06	.34	.50	.19	-18

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 201
WASHER, FINISHING, COUNTERSUNK
APPLICABLE DOCUMENT: MS27129

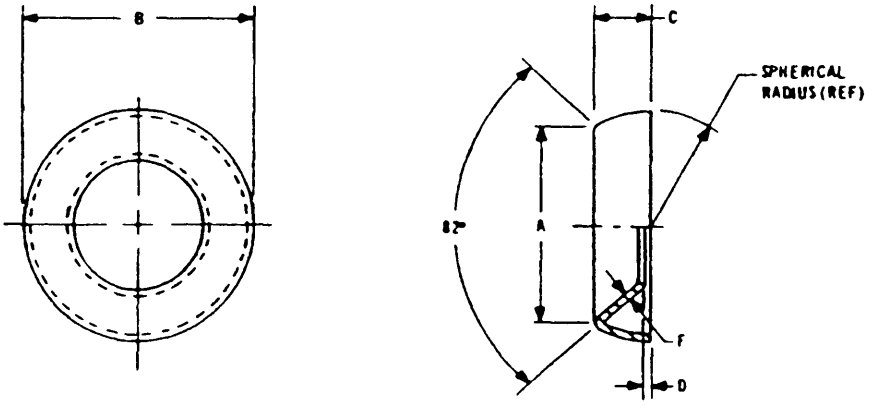


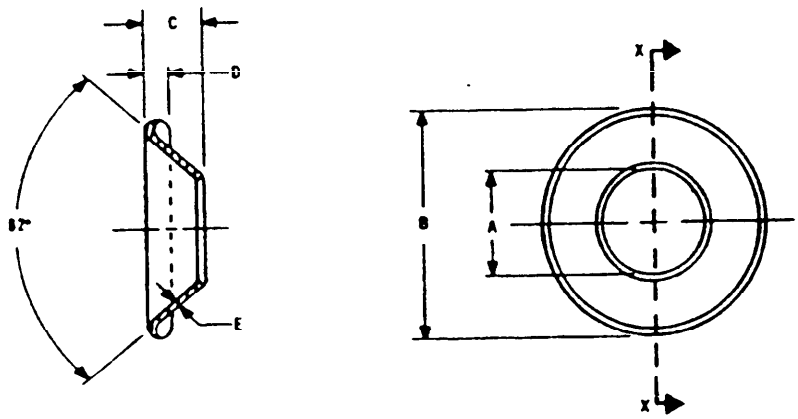
TABLE I. Part numbers.

Material						Brass	Steel	CRES
Protective finish						Chromium or nickel plate	Cadmium plate	Passivate
Hardness (BHN) min						104		
Size	A Max	B Max	C Max	D Max	F Min	MS27129 + Dash no.		
.112	.247	.385	.206	.036	.012	-1	-2	-3
.138	.303	.448	.102	.036	.014	-4	-5	-6
.164	.357	.541	.112	.036	.014	-7	-8	-9
.190	.437	.603	.122	.036	.018	-10	-11	-12
.250	.546	.791	.170	.051	.018	-16	-17	-18
.3125	.678	.947	.222	.051	.020	-19	-20	-21
.375	.810	1.135	.260	.051	.025	-22	-23	-24

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 202
WASHER, FINISHING, FLUSH
APPLICABLE DOCUMENT: MAS390



SECTION X-X

TABLE I. Part numbers.

Material	Carbon steel	Brass
Protective finish	Cadmium plate	Chromium or nickel plate
A Min / B Max / C Max / D Max / E Max	MAS390 - Dash no. 1/	
.120 / .320 / .093 / .046 / .016	4	
.150 / .390 / .100 / .046 / .018	6	
.180 / .443 / .113 / .046 / .018	8	
.193 / .500 / .135 / .046 / .022	10	
.252 / .650 / .162 / .061 / .022	14	

1/ Add "B" in front of dash number for brass.
Add "S" in front of dash number for steel.
Add dash in place of code letter when material is optional.
Add "P" after dash number for chromium or nickel plate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 301
WASHER, FLAT, RECTANGULAR
APPLICABLE DOCUMENT MS51867

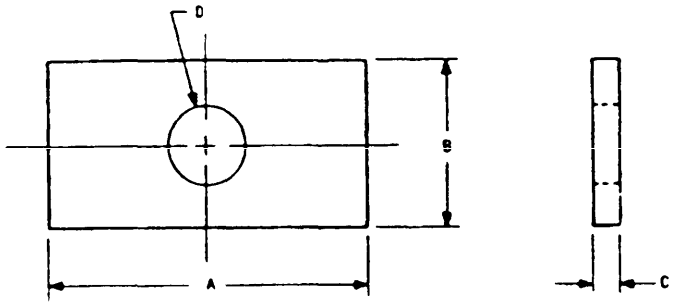


TABLE I. Part numbers.

Material				Carbon steel	CRES	Aluminum
Protective finish				Cadmium plate	Passivate	Anodize
D Min	A	B	C Max	MS51867 + Dash no.		
.120	.375	.312	.040	-S1	-C1	-A1
.151	.438	.375	.065	-S2	-C2	-A2
.183	.500	.438	.065	-S3	-C3	-A3
.214	.625	.500	.065	-S4	-C4	-A4
.276	1.000	.625	.080	-S6	-C6	-A6
.339	1.125	.688	.080	-S7	-C7	-A7
.401	1.500	.812	.080	-S8	-C8	-A8
.526	2.000	1.062	.121	-S9	-C9	-A9
.649	2.500	1.312	.121	-S10	-C10	-A10
.805	2.750	1.469	.160	-S11	-C11	-A11
1.055	3.500	2.000	.192	-S12	-C12	-A12
1.368	5.000	2.500	.192	-S13	-C13	-A13
1.618	6.000	3.000	.192	-S14	-C14	-A14
1.865	8.000	4.000	.213	-S15	-C15	-A15
2.115	9.000	4.500	.213	-S16	-C16	-A16

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE I. Part numbers (aluminum alloy). - Continued

Material			Aluminum alloy					
			5052				7075	
Protective finish			None				Anodize	
A Min	B Max	C	MAS620 - Dash no	MAS1197 - Dash no	AM960 - Dash no		MAS1252 - Dash no	MS15795 - Dash no
.443	.770	.016	--	-716L	D716L	PD716L	--	--
		.032	--	--	--	--	-716L	--
		.063	--	-716	D716	PD716	-716M	--
.495	1.280	.083	--	--	--	--	--	-717
.505	.895	.016	--	-816L	D816L	PD816L	--	--
		.032	--	--	--	--	-816L	--
		.063	--	-816	D816	PD816	-816M	--
.630	1.208	.016	--	-1016L	D1016L	PD1016L	--	--
		.032	--	--	--	--	-1016L	--
		.063	--	-1016	D1016	PD1016	-1016M	--
.681	1.780	.134	--	--	--	--	--	-721
.755	1.332	.016	--	-1216L	D1216L	PD1216L	--	--
		.032	--	--	--	--	-1216L	--
		.063	--	--	--	--	-1216M	--
		.090	--	-1216	D1216	PD1216	--	--
.805	1.499	.134	--	--	--	--	--	-722
	2.030	.148	--	--	--	--	--	-723
.880	1.520	.016	--	-1416L	D1416L	PD1416L	--	--
		.032	--	--	--	--	-1416L	--
		.063	--	--	--	--	-1416M	--
		.090	--	-1416	D1416	PD1416	--	--
.931	1.780	.134	--	--	--	--	--	-724
	2.280	.165	--	--	--	--	--	-725
1.005	1.770	.016	--	-1616L	D1616L	PD1616L	--	--
		.032	--	--	--	--	-1616L	--
		.063	--	--	--	--	-1616M	--
		.090	--	-1616	D1616	PD1616	--	--
1.130	1.895	.016	--	--	D1816L	PD1816L	--	--
		.032	--	--	--	--	-1816L	--
		.063	--	--	--	--	-1816M	--
		.090	--	--	D1816	PD1816	--	--
1.243	2.780	.165	--	--	--	--	--	-728
1.255	2.020	.016	--	--	D2016L	PD2016L	--	--
		.032	--	--	--	--	-2016L	--
		.063	--	--	--	--	-2016M	--
		.090	--	--	D2016	PD2016	--	--
1.368	3.030	.165	--	--	--	--	--	-729
1.490	3.295	.180	--	--	--	--	--	-730
1.630	2.395	.016	--	--	D2516L	PD2516L	--	--
		.032	--	--	--	--	-2516L	--
		.063	--	--	--	--	-2516M	--
		.090	--	--	D2516	PD2516	--	--
1.740	3.795	.180	--	--	--	--	--	-732
1.880	2.645	.016	--	--	D3016L	PD3016L	--	--
		.032	--	--	--	--	-3016L	--
		.063	--	--	--	--	-3016M	--
		.090	--	--	D3016	PD3016	--	--
1.990	4.295	.180	--	--	--	--	--	-734
2.255	3.020	.016	--	--	D3616L	PD3616L	--	--
		.032	--	--	--	--	-3616L	--
		.063	--	--	--	--	-3616M	--
		.090	--	--	D3616	PD3616	--	--
2.365	4.795	.220	--	--	--	--	--	-736
2.505	3.270	.016	--	--	D4016L	PD4016L	--	--
		.032	--	--	--	--	-4016L	--
		.063	--	--	--	--	-4016M	--
		.090	--	--	D4016	PD4016	--	--
2.615	5.045	.238	--	--	--	--	--	-737
2.865	5.315	.258	--	--	--	--	--	-738
3.115	5.565	.284	--	--	--	--	--	-739

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 302
WASHER, FLAT, ROUND
APPLICABLE DOCUMENTS: NAS549, NAS620, NAS1197, NAS1252, NAS1515, NAS1587, AN960, AN961, AN970, MS9320, MS9321, MS9549, MS14151, MS15795, MS16212, MS20002, MS21206, MS25440, MS27183, MS51859

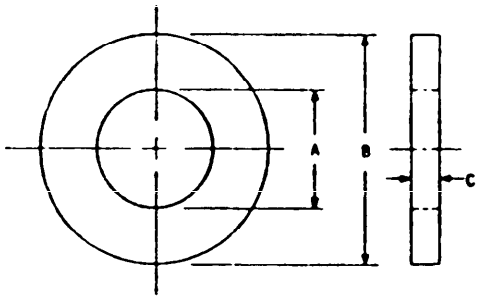


TABLE I. Part numbers (aluminum alloy).

Material			Aluminum alloy					
			5052				7075	
Protective finish			None				Anodize	
A Min	B Max	C	NAS620 - Dash no.	NAS1197 - Dash no.	AN960 - Dash no.		NAS1252 - Dash no.	MS15795 - Dash no.
.063 .089 .089	.114 .164 .250	.016 .016 .020	A0 A2 —	— — —	— — —	— — —	— — —	— — -702
.095	.270	.016 .032	— —	-2L -2	— —	— —	— —	— —
.115	.224	.016 .032	A4L A4	— —	— —	— —	— —	— —
.120	.250	.022	—	—	—	—	—	-703
.115 .120	.332	.016 .032	— —	-4L -4	D4L D4	PD4L PD4	— -4L	— -704
.143	.282	.016 .032	A6L A6	— —	— —	— —	— —	— —
.139	.395	.016 .032	— —	-6L -6	D6L D6	PD6L PD6	— -6L	— —
.164	.395	.016 .032	— —	-8L -8	D8L D8	PD8L PD8	— -8L	— —
.169	.319	.016 .032	A8L A8	— —	— —	— —	— —	— —
.183	.390 .453	.049 .049	— —	— —	— —	— —	— —	-707 -741
.193	.458	.016 .032 .063	— — —	-10L — -10	D10L — D10	PD10L — PD10	— -10L -10H	— — —
.195	.369	.032 .063	A10L A10	— —	— —	— —	— —	— —
.245	.577	.065	—	—	—	—	—	-709
.255	.483	.032 .063	A416L A416	— —	— —	— —	— —	— —
.255	.520	.016 .032 .063	— — —	-416L — -416	D416L — D416	PD416L — PD416	— -416L -416H	— — —
.307	.749	.065	—	—	—	—	—	-711
.318	.582	.016 .032 .063	— — —	-516L — -516	D516L — D516	PD516L — PD516	— -516L -516H	— — —
.370	.905	.083	—	—	—	—	—	-713
.380	.645	.016 .032 .063	— — —	-616L — -616	D616L — D616	PD616L — PD616	— -616L -616H	— — —
.433	1.030	.083	—	—	—	—	—	-715

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE II. Part numbers (alloy steel).

Material			Alloy steel		
Protective finish			Nickel-cadmium plate	Cadmium plate	
Hardness (Rockwell)			C35-40	—	C38-45
Tensile strength (psi)			—	125,000 to 145,000	—
A Min	B Max	C	MS9320 - Dash no.	MS20002 - Dash no.	MS21206 - Dash no.
.130	.322	.031	-05	—	—
.154	.385	.043	-07	—	—
.179	.385	.050	-08	—	—
.191	.372	.062	—	—	-3
.208	.448	.062	-09	—	—
.251	.470	.062	—	—	-4
.252	.541	.062	—	-4	—
.271	.510	.062	-10	—	—
.314	.573	.062	—	—	-5
.315	.603	.062	—	-5	—
.333	.635	.062	-11	—	—
.376	.679	.062	—	—	-6
.378	.697	.062	—	-6	—
.396	.760	.078	-12	—	—
.439	.783	.062	—	—	-7
.441	.791	.062	—	-7	—
.458	.885	.094	-13	—	—
.501	.898	.062	—	—	-8
.504	.885	.062	—	-8	—
.521	1.010	.109	-14	—	—
.627	1.125	.062	—	—	-10
.631	1.072	.062	—	-10	—
.646	1.260	.141	-16	—	—
.752	1.350	.062	—	—	-12
.757	1.260	.062	—	-12	—
.771	1.510	.172	-17	—	—
.877	1.572	.062	—	—	-14
.884	1.447	.062	—	-14	—
.896	1.760	.203	-18	—	—
1.002	1.798	.062	—	—	-16
1.010	1.635	.062	—	-16	—
1.021	2.010	.234	-19	—	—
1.127	2.188	.062	—	—	-18
1.135	1.885	.062	—	-18	—
1.252	2.260	.062	—	—	-20
1.260	2.135	.062	—	-20	—
1.377	2.490	.062	—	—	-22
1.385	2.323	.062	—	-22	—
1.502	2.723	.062	—	—	-24
1.510	2.510	.062	—	-24	—

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE III. Part numbers (carbon steel).

Material			Carbon steel				
Protective finish			Cadmium plate				
A Min	B Max	C	AN960 - Dash no.	AN970 - Dash no.	NAS620 - Dash no.	MS25400 - Dash no.	MS27183 - Dash no.
.063	.114	.016	--	--	-0	--	--
.089	.164	.016	--	--	-2	--	--
.089	.250	.020	--	--	--	--	-2
.115	.224	.016	--	--	-4L	--	--
		.032	--	--	-4	--	--
.115	.332	.016	-4L	--	--	--	--
		.032	-4	--	--	--	--
.120	.250	.022	--	--	--	--	-3
.120	.320	.032	--	--	--	--	-4
.139	.395	.016	-6L	--	--	--	--
		.032	-6	--	--	--	--
.143	.282	.016	--	--	-6L	--	--
		.032	--	--	-6	--	--
.164	.395	.016	-8L	--	--	--	--
		.032	-8	--	--	--	--
.169	.319	.016	--	--	-8L	--	--
		.032	--	--	-8	--	--
.183	.390	.049	--	--	--	--	-7
	.453	.049	--	--	--	--	-41
.193	.688	.074	--	--	--	-3	--
.193	.458	.032	-10L	--	--	--	--
		.063	-10	--	--	--	--
	.885	.063	--	-3	--	--	--
.195	.369	.032	--	--	-10L	--	--
		.063	--	--	-10	--	--
.245	.575	.065	--	--	--	--	-9
.255	.483	.032	--	--	-416L	--	--
		.063	--	--	-416	--	--
.255	.520	.032	-416L	--	--	--	--
		.063	-416	--	--	--	--
	1.135	.063	--	-4	--	--	--
.260	.688	.093	--	--	--	-4	--
	.875	.093	--	--	--	-4A	--
.307	.749	.065	--	--	--	--	-11
.318	.582	.032	-516L	--	--	--	--
		.063	-516	--	--	--	--
	1.385	.063	--	-5	--	--	--
.320	.875	.093	--	--	--	-5	--
.370	.905	.083	--	--	--	--	-13
.380	.645	.032	-616L	--	--	--	--
		.063	-616	--	--	--	--
	1.635	.063	--	-6	--	--	--
.385	.875	.125	--	--	--	-6	--
	1.187	.125	--	--	--	-6A	--
.433	1.030	.083	--	--	--	--	-15
.443	.770	.032	-716L	--	--	--	--
		.063	-716	--	--	--	--
	1.822	.109	--	-7	--	--	--
.495	1.280	.083	--	--	--	--	-17
.510	1.187	.125	--	--	--	-8	--
.505	.895	.032	-816L	--	--	--	--
		.063	-816	--	--	--	--
	2.010	.109	--	-8	--	--	--
.630	1.208	.032	-1016L	--	--	--	--
		.063	-1016	--	--	--	--
	2.385	.125	--	-10	--	--	--
.681	1.788	.134	--	--	--	--	-22

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the currentversion before use.

TABLE III. Part numbers (carbon steel). - Continued

Material			Carbon steel				
Protective finish			Cadmium plate				
A Min	B Max	C	AN960 - Dash no.	AN970 - Dash no.	NAS620 - Dash no.	MS25440 - Dash no.	MS27183 - Dash no.
.755	1.332	.032 .090	-1216L -1216	— —	— —	— —	— —
.805	1.499 2.030	.134 .148	- -	— —	— —	— —	-23 -24
.880	1.520	.032 .090	-1416L -1416	— —	— —	— —	— —
.931	1.780 2.280	.134 .165	— —	— —	— —	— —	-25 -26
1.005	1.770	.032 .090	-1616L -1616	— —	— —	— —	— —
1.130	1.895	.032 090	-1816L -1816	— —	— —	— —	— —
1.243	2.780	.165	-	—	—	—	-29
1.255	2.020	.032 .090	-2016L -2016	— —	— —	— —	— —
1.368	3.030	.165	-	—	—	—	-30
1.490	3.295	.180	-	—	—	—	-31
1.630	2.395	.032 .090	-2616L -2616	— —	— —	— —	— —
1.740	3.795	.180	-	—	—	—	-33
1.880	2.645	.032 .090	-3016L -3016	— —	— —	— —	— —
1.990	4.295	.180	-	—	—	—	-35
2.255	3.020	.032 .090	-3616L -3616	— —	— —	— —	— —
2.365	4.795	.220	-	—	—	—	-37
2.505	3.270	.032 .090	-4016L -4016	— —	— —	— —	— —
2.615	5.045	.238	-	—	—	—	-38
2.865	5.315	.259	-	—	—	—	-39
3.115	5.565	.284	-	—	—	—	-40

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE IV. Part numbers (CRES).

Material			CRES							
Protective finish			Black oxide	Uncoated		Passivate				
Tensile strength (psi) min.			Not specified						75,000	90,000
A Min	B Max	C	AMS960 - Dash no.	MS9321 - Dash no.	MS9549 - Dash no.	NAS620 - Dash no.	MS14151 - Dash no.	MS15795 - Dash no.	NAS1587 - Dash no.	MS16212 - Dash no.
.063 .089 .089	.114 .164 .250	.016 .016 .020	— — —	— — —	— — —	C10 C12 —	— — —	— — -802	— — —	— — —
.115	.224	.016 .032	— —	— —	— —	C4L C4	— —	— —	— —	— —
.115	.332	.016 .032	XC4L XC4	— —	— —	— —	— —	— —	— —	— —
.120	.250	.022	—	—	—	—	—	-803	—	—
.130	.322	.016 .031	— —	— -05	-05 —	— —	— —	— —	— —	— —
.139	.395	.016 .032	XC6L XC6	— —	— —	— —	— —	— —	— —	— —
.143	.282	.016 .032	— —	— —	— —	C6L C6	— —	— —	— —	— —
.144	.490	.020	—	—	—	—	-1	—	—	—
.154	.385	.016 .043	— —	— -07	-07 —	— —	— —	— —	— —	— —
.164	.395	.016 .032	XC8L XC8	— —	— —	— —	— —	— —	— —	— —
.167	.560	.020	—	—	—	—	-6	—	—	—
.169	.319	.016 .032	— —	— —	— —	C8L C8	— —	— —	— —	— —
.179	.385	.016 .032	— —	— -08	-08 —	— —	— —	— —	— —	— —
.183	.390 .453	.049 .049	— —	— —	— —	— —	— —	-807 -841	— —	— —
.192	.479	.032	—	—	—	—	—	—	-3	—
.193	.458	.032 .063	XC10L XC10	— —	— —	— —	— —	— —	— —	— —
.195	.369	.032 .063	— —	— —	— —	C10L C10	— —	— —	— —	— —
.198	.620	.020	—	—	—	—	-2	—	—	—
.203	.448	.032 .062	— —	— -09	-09 —	— —	— —	— —	— —	— —
.245	.577	.065	—	—	—	—	—	-809	—	—
.252	.541	.062	—	—	—	—	—	—	-4	—
.255	.483	.032 .063	— —	— —	— —	C416L C416	— —	— —	— —	— —
	1.090	.020	—	—	—	—	-3	—	—	—
.255	.520	.032 .063	XC416L XC416	— —	— —	— —	— —	— —	— —	— —
.271	.510	.032 .062	— —	— -10	-10 —	— —	— —	— —	— —	— —
.302 .307	.760 .758	.080 .065	— —	— —	— —	— —	— —	— -811	— —	-11 —
.315 .318	603 1.090	.062 .020	— —	— —	— —	— —	— -4	— —	-5 —	— —

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE IV Part numbers (CRES) - Continued

Material			CRES								
Protective Finish			Black oxide	Uncoated		Passivate					
Tensile strength (psi) min			Not specified							75,000	50,000
A Min	B Max	C	AN960 - Dash no.	MS9321 - Dash no.	MS9549 - Dash no.	NAS620 - Dash no.	MS14151 - Dash no.	MS15795 - Dash no.		NAS1587 - Dash no.	MS16212 - Dash no.
.318	.582	.032 .063	XC516L XC516	— —	— —	— —	— —	— —		— —	— —
.333	.572 .635	.032 .062	— —	— -11	-11 —	— —	— —	— —		— —	— —
.365 .370	.760 .905	.080 .083	— —	— —	— —	— —	— —	— -813		— —	-12 —
.378 .380	.697 1.090	.062 .020	— —	— —	— —	— —	— -5	— —		-6 —	— —
.380	.645	.032 .063	XC616L XC616	— —	— —	— —	— —	— —		— —	— —
.386	.635 .760	.032 .078	— —	— -12	-12 —	— —	— —	— —		— —	— —
.427 .433	.885 1.030	.104 .083	— —	— —	— —	— —	— —	— -815		— —	-13 —
.441	.791	.062	—	—	—	—	—	—		-7	—
.443	.770	.032 .063	XC716L XC716	— —	— —	— —	— —	— —		— —	— —
.458	.760 .885	.032 .094	— —	— -13	-13 —	— —	— —	— —		— —	— —
.495 .504	1.280 .885	.083 .062	— —	— —	— —	— —	— —	-817 —		— -8	— —
.505	.895	.032 .063	XC816L XC816	— —	— —	— —	— —	— —		— —	— —
.521	.802 1.010	.032 .109	— —	— -14	-14 —	— —	— —	— —		— —	— —
.552 .631	1.260 1.072	.132 .062	— —	— —	— —	— —	— —	— —		— -10	-14 —
.630	1.208	.032 .063	XC1016L XC1016	— —	— —	— —	— —	— —		— —	— —
.646	1.010 1.260	.032 .141	— —	— -16	-16 —	— —	— —	— —		— —	— —
.677	1.510	.160	—	—	—	—	—	—		—	-15
.681 .757	1.780 1.260	.134 .062	— —	— —	— —	— —	— —	-821 —		— -12	— —
.755	1.332	.032 .090	XC1216L XC1216	— —	— —	— —	— —	— —		— —	— —

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE IV Part numbers (CRES) - Continued

Material			CRES							
Protective finish			Black oxide	Uncoated		Passivate				
Tensile strength (psi) min.			Not specified						75,000	50,000
A Min	B Max	C	AN960 - Dash no.	MS9321 - Dash no.	MS9549 - Dash no.	NAS620 - Dash no.	MS14151 - Dash no.	MS15795 - Dash no.	NAS1587 - Dash no.	MS16212 - Dash no.
.771	1.198 1.510	.032 .172	-- —	-- -17	-17 —	-- --	-- --	— --	— --	— --
.802	1.760	.177	--	--	--	--	--	—	--	-16
.805	1.499 2.000	.134 .148	— --	-- --	-- --	-- --	-- --	-822 -823	-- --	-- --
.880	1.520	.032 .090	XC1416L XC1416	-- --	-- --	-- --	-- --	— --	— —	— —
.884	1.447	.062	—	--	--	--	—	—	-14	--
.896	1.385 1.760	.032 .203	— —	— -18	-18 --	— —	— —	— —	— —	— —
.927	2.010	.192	—	—	--	—	--	—	—	-17
.931	1.750 2.250	.134 .165	— —	— --	— —	— --	— --	-824 -835	— --	— --
1.010	1.635	.062	—	—	—	—	—	—	-16	--
1.005	1.770	.032 .090	XC1616L XC1616	— —	— —	— —	— —	— —	— —	— —
1.021	1.572 2.018	.032 .234	— --	— -19	-19 —	— --	— --	— —	— —	— —
1.052	2.260	.192	—	—	—	—	—	—	—	-18
1.130 1.135 1.243	1.895 1.885 2.730	.090 .062 .165	XC1816 — —	— — —	— — —	— — —	— — —	— — -828	— -18 —	— — —
1.255 1.260 1.302	2.028 2.730 3.010	.090 .062 .213	XC2016 — —	— — —	— — —	— — —	— — —	— — —	— -20 —	— — -19
1.368 1.490 1.552	3.030 3.295 3.260	.165 .180 .213	— — —	— — —	— — —	— — —	— — —	-829 -830 —	— — —	— — -20
1.630 1.740 1.880	2.395 3.795 2.645	.090 .180 .090	XC2616 — XC3016	— — —	— — —	— — —	— — —	— -832 —	— — —	— — —
1.990 2.255 2.365	4.295 3.020 4.795	.180 .090 .220	— XC3616 —	— — —	— — —	— — —	— — —	-834 — -836	— — —	— — —
2.505 2.615 2.865 3.115	3.270 5.045 5.565 5.315	.090 .238 .259 .284	XC4016 — — —	— — — —	— — — —	— — — —	— — — —	— -837 -838 -839	— — — —	— — — —

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE V Part numbers (brass)

Material			Brass			
Protective finish			Not specified	Cadmium plate	Tin or silver plate	Black oxide
A Min	B Max	C	AN960 - Dash no	NAS620 - Dash no	AN961 - Dash no. 1/	MS15795 - Dash no.
.062	.114	.016	--	B0	--	--
.089	.164	.016	--	B2	--	--
.089	.250	.020	--	--	--	-902
.115	.224	.016	--	B4L	--	--
		.032	--	B4	--	--
.115	.332	.032	B4	--	-4	--
.170	.250	.022	--	--	--	-903
	.320	.032	--	--	--	-904
.139	.395	.032	B6	--	-6	--
.143	.282	.016	--	B6L	--	--
		.032	--	B6	--	--
.164	.395	.032	B8	--	-8	--
.169	.319	.016	--	B8L	--	--
		.032	--	B8	--	--
.183	.390	.049	--	--	--	-907
	453	.049	--	--	--	-941
.193	.458	.032	--	--	-10	--
		.063	B10	--	--	--
.195	.369	.032	--	B10L	--	--
		.063	--	B10	--	--
.245	.577	.065	--	--	--	-909
.255	.483	.032	--	B416L	--	--
		.063	--	B416	--	--
.255	.520	.063	B416	--	--	--
		.064	--	--	-416	--
.307	.749	.065	--	--	--	-911
.318	.582	.063	B516	--	--	--
		.064	--	--	-516	--
.330	.645	.063	B616	--	--	--
		.064	--	--	-616	--
.370	.905	.083	--	--	--	-913
.433	1.030	.083	--	--	--	-915
.443	.770	.063	B716	--	--	--
.495	1.280	.083	--	--	--	-917
.505	.895	.063	B816	--	--	--
		.064	--	--	-816	--
.630	1.208	.063	B1016	--	--	--
		.064	--	--	-1016	--
.681	1.780	.134	--	--	--	-921
.755	1.332	.090	B1216	--	--	--
.805	1.499	.134	--	--	--	-922
	2.030	.148	--	--	--	-923

1/ Add "S" after dash number for silver plate.

Add "T" after dash number for tin plate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE V. Part numbers (brass) - Continued

Material			Brass			
Protective finish			Not specified	Cadmium plate	Tin or silver plate	Black oxide
A Min	B Max	C	AN960 - Dash no.	MAS620 - Dash no.	AN961 - Dash no 1/	MS15795 - Dash no.
.880	1.520	.090	B1416	—	—	--
.931	1.780	.134	—	--	—	-924
	1.280	.165	—	--	--	-925
1.005	1.770	.090	B1616	—	—	—
1.130	1.895	.090	B1816	—	—	--
1.243	2.780	.165	--	--	--	-928
1.255	2.020	.090	B2016	--	—	—
1.368	3.030	.165	--	--	--	-929
1.490	3.295	.180	--	--	—	-930
1.630	2.395	.090	B2616	--	--	--
1.740	3.795	.180	--	--	—	-932
1.880	2.645	.090	B3016	—	—	—
1.990	4.250	.180	—	--	--	-934
2.255	3.020	.090	B3616	—	—	—
2.365	4.795	.220	--	--	--	-936
2.505	3.270	.090	B4016	--	—	—
2.615	5.045	.238	—	—	--	-937
2.865	5.315	.259	—	--	--	-938
3.115	5.565	.284	--	--	—	-939

1/ Add "S" after dash number for silver plate.

Add "T" after dash number for tin plate.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE VI Part numbers (copper alloys).

Material			Copper	Nickel-copper alloy	Copper-silicon alloy
Protective finish			Uncoated		Passivate
Tensile strength (psi) min			Not specified		50,000
A Min	B Max	C	MS15795 - Dash no.		MS16212 - Dash no.
.089	.250	020	-502	-402	--
.129	.250 .320	022 032	-503 -504	-403 -404	-- --
.182	.442	.065	—	--	-21
.183	.390 .453	049 049	-507 -541	-407 -441	-- --
.215	577	065	-509	-409	--
.245	567	065	--	--	-22
.302	.760	.080	--	--	-1
.307	.749	.065	-511	-411	--
.365	.760	.080	—	--	-2
.370	.905	.083	-513	-413	--
.427	.885	.104	—	—	-3
.433	1.030	.083	-515	-415	--
.495	1.280	.083	-517	-417	—
.552	1.260	.132	--	--	-4
.677	1.510	.160	--	--	-5
.681	1.780	.134	-521	-421	—
.802	1.760	.177	--	—	-6
.805	1.469	.134	-522	-422	--
.805	2.030	.148	-523	-423	--
.927	2.010	.192	--	—	-7
.931	1.780 2.280	.134 .165	-524 -425	-424 -425	-- --
1.052	2.260	.192	--	--	-8
1.243	2.780	.165	-528	-428	—
1.302	3.010	.213	—	--	-9
1.368	3.030	.165	-529	-429	—
1.490	3.295	.180	-530	-430	--
1.552	3.260	.213	--	—	-10
1.740	3.795	.180	-532	-432	—
1.990	4.295	.180	-534	-434	--
2.365	4.795	.220	-536	-436	--
2.615	5.045	.238	-537	-437	--
2.865	5.315	.259	-538	-438	--
3.115	5.565	.284	-539	-439	--

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE VII Part numbers (non-metallic material)

Material			1/	2/	Nylon
Protective finish			None		
A Min	B Max	C	NAS549 - Dash no	NAS1515 - Dash no	MS51859 - Dash no
.090	.203	.031	—	—	-1
.100	.770	.031 .063	L3 3	02L 02	— —
.115	.297	.047	—	—	-2
.120	.332	.031 .063	L4 4	04L 04	— —
.140	.328	.047	—	—	-3
.144	.395	.031 .063	L6 6	06L 06	— —
.169	.395	.031 .063	L8 8	08L 08	— —
.170	.406	.047	—	—	-4
.198	.458	.031 .063 .094	L10 10 H10	3L 3 3H	— — —
.200	.469	.047	—	—	-5
.255	.520	.031 .063 .094	L416 416 H416	4L 4 4H	— — —
.260	.688	.047	—	—	-6
.318	.582	.031 .063 .094	L516 516 H516	5L 5 5H	— — —
.334	.708	.047	—	—	-7
.380	.645	.031 .063 .094	L616 616 H616	6L 6 6H	— — —
.396	.832	.047	—	—	-8
.443	.770	.031 .063 .094	L716 716 H716	7L 7 7H	— — —
.459	.942	.047	—	—	-9
.505	.895	.031 .063 .094	L816 816 H816	8L 8 8H	— — —
.521	1.082	.047	—	—	-10
.630	1.208	.031 .063 .094	L1016 1016 H1016	10L 10 10H	— — —
.755	1.332	.031 .063 .094	L1216 1216 H1216	12L 12 12H	— — —
.880	1.520	.031 .063 .094	L1416 1416 H1416	14L 14 14H	— — —
1.005	1.770	.031 .063 .094	L1616 1616 H1616	16L 16 16H	— — —

1/ NAS 549

Letter before dash number indicates type of material

- F Fiber, insulation
- P Phenolic - paper
- M Phenolic - cotton (mechanical grade)
- B Phenolic - cotton (general purpose grade)
- E Phenolic cotton - (electrical grade)
- P Melamine - glass
- G Epoxy - glass

2/ NAS 1515

Letter before dash number indicates type of material

- A Silicone - low temp (-100°F)
- B Silicone - high temp. (450°)
- C Syn rubber - fuel resistant
- D Syn rubber - oil resistant
- E Syn rubber - oil resistant for use with acrylic
- F Buna N-diester syn oil resistant
- H Nylon
- J Chloroprene - weather resistant
- K Butyl - inorganic ester resistant
- L Kel-F-temp max 350°F, volt/mil 2800
- M Teflon-temp max 500°F, volt/mil 350

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 303
WASHER, FLAT, SQUARE
APPLICABLE DOCUMENT: MS21307

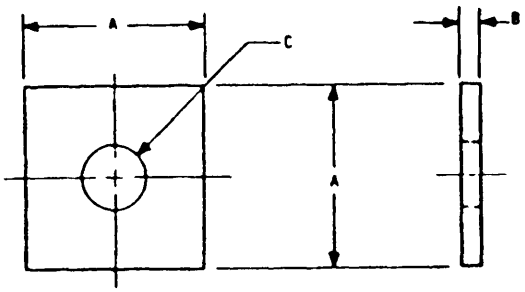


TABLE I Part numbers.

Material			Steel
Protective finish			Galvanized
C	A	B	MS21307 - Dash no.
1 1/16	2	1/8	-1
	2-1/4	3/16	-2
1 3/16	2-1/4	3/16	-3
	3	3/16	-4
		1/4	-5
1 5/16	3-1/2	3/8	-6
1-1/8	4	1/2	-7

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 304
WASHER, FLAT, SQUARE WITH RADIUS
APPLICABLE DOCUMENT: NAS1401

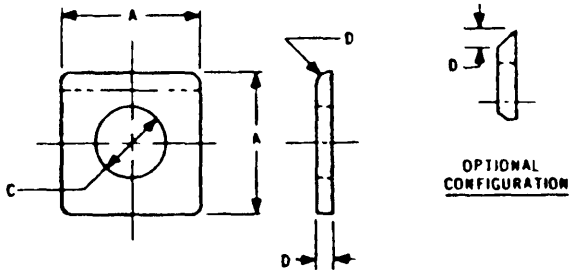


TABLE I. Part numbers.

Material			Aluminum alloy	Carbon steel	CRES
Protective finish			Anodize	Cadmium plate	Passivate
C Min	A Max	D	NAS1401 - Dash no.		
.193	.490	.090 .160	-3D3 -3D5	-3-3 -3-5	-3C3 -3C5
.256	.570	.090 .160	-4D3 -4D5	-4-3 -4-5	-4C3 -4C5
.318	.640	.090 .160	-5D3 -5D5	-5-3 -5-5	-5C3 -5C5
.380	.710	.090 .160 .250	-6D3 -6D5 -6D8	-6-3 -6-5 -6-8	-6C3 -6C5 -6C8
.443	.780	.090 .160 .250	-7D3 -7D5 -7D8	-7-3 -7-5 -7-8	-7C3 -7C5 -7C8
.505	.930	.090 .160 .250	-8D3 -8D5 -8D8	-8-3 -8-5 -8-8	-8C3 -8C5 -8C8
.630	1.150	.090 .160 .250	-10D3 -10D5 -10D8	-10-3 -10-5 -10-8	-10C3 -10C5 -10C8
.755	1.290	.090 .160 .250	-12D3 -12D5 -12D8	-12-3 -12-5 -12-8	-12C3 -12C5 -12C8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 401
WASHER KEY, BEARING
APPLICABLE DOCUMENTS MS19069, MS19070, MS17277 thru MS17328

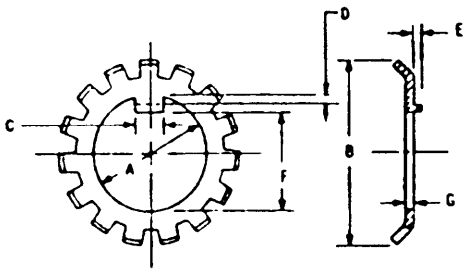


TABLE I. Part numbers.

Material							Steel	Carbon steel	Carbon steel	CRES	Alloy steel
Protective finish							Phosphate	None	Zinc or Cd plate	Passivate	Phosphate
A Min	B Max	C Max	D Max	E Min	F Min	G Nom	Part number	MS19070 - Dash no.			MS19069 - Dash no.
.406	.828	.120	.030	.062	--	.042	--	--	--	--	-A00
.406	.875	.120	.030	.062	--	.042	--	-001	-002	-003	--
.484	.922	.120	.030	.062	--	.042	--	--	--	--	-A01
.484	1.016	.120	.030	.062	--	.042	--	-011	-012	-013	--
.515	1.156	.120	--	.062	.443	.040	MS17277I	--	--	--	--
.601	1.094	.120	.030	.062	--	.042	--	--	--	--	-A02
.601	1.156	.120	.030	.062	--	.042	--	-021	-022	-023	--
.641	1.312	.120	--	.062	.567	.040	MS17277J	--	--	--	--
.679	1.203	.120	.030	.062	--	.042	--	--	--	--	-A03
.679	1.328	.120	.030	.062	--	.042	--	-031	-032	-033	--
.770	1.578	.176	--	.062	.677	.040	MS17277S	--	--	--	--
.801	1.344	.176	.030	.062	--	.042	--	--	--	--	-A04
.801	1.531	.176	.030	.062	--	.042	--	-041	-042	-043	--
.895	1.719	.176	--	.094	.802	.050	MS17277T	--	--	--	--
.989	1.562	.176	.030	.094	--	.050	--	--	--	--	-A05
.989	1.719	.176	.030	.094	--	.050	--	-051	-052	-053	--
1.020	1.891	.176	--	.094	.927	.050	MS17277U	--	--	--	--
1.145	2.047	.176	--	.094	1.052	.050	MS17277V	--	--	--	--
1.193	1.859	.176	.050	.094	--	.050	--	--	--	--	-A06
1.193	1.922	.176	.050	.094	--	.050	--	-061	-062	-063	--
1.270	2.172	.176	--	.094	1.177	.050	MS17277W	--	--	--	--
1.333	2.750	.176	.050	.094	--	.050	--	-0651	-0652	-0653	--
1.395	2.297	.176	--	.094	1.302	.050	MS17277X	--	--	--	--
1.396	2.078	.176	.050	.094	--	.050	--	--	--	--	-A07
1.396	2.250	.176	.050	.094	--	.050	--	-071	-072	-073	--
1.520	2.406	.239	--	.094	1.411	.063	MS17277Y	--	--	--	--
1.583	2.250	.300	.050	.094	--	.058	--	--	--	--	-A08
1.583	2.469	.290	.050	.094	--	.058	--	-081	-082	-083	--
1.775	2.719	.239	--	.125	1.661	.063	MS17277Z	--	--	--	--
1.792	2.500	.300	.050	.125	--	.058	--	--	--	--	-A09
1.792	2.734	.290	.050	.125	--	.058	--	-091	-092	-093	--
1.992	2.688	.300	.050	.125	--	.058	--	--	--	--	-A10
1.992	2.922	.290	.050	.125	--	.058	--	-101	-102	-103	--
2.025	2.969	.239	--	.125	1.911	.063	MS17278A	--	--	--	--
2.182	2.953	.300	.050	.125	--	.063	--	--	--	--	-A11
2.182	3.109	.290	.050	.125	--	.063	--	-111	-112	-113	--
2.281	3.234	.239	--	.125	2.161	.063	MS17278B	--	--	--	--
2.400	3.188	.300	.070	.125	--	.063	--	--	--	--	-A12
2.400	3.344	.290	.070	.125	--	.063	--	-121	-122	-123	--
2.531	3.484	.239	--	.125	2.411	.063	MS17278C	--	--	--	--
2.588	3.375	.300	.070	.125	--	.063	--	--	--	--	-A13
2.588	3.578	.290	.070	.125	--	.063	--	-131	-132	-133	--

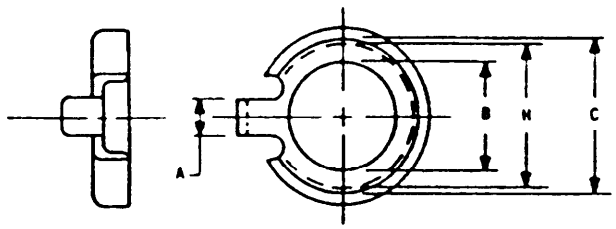
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE I Part numbers. - Continued

Material							Steel	Carbon steel	Carbon steel	CRES	Alloy steel
Protective finish							Phosphate	None	Zinc or cd plate	Passivate	Phosphate
A Min	B Max	C Max	D Max	E Min	F Min	G Nom	Part number	MS19070 - Dash no.			MS19069 - Dash no.
2.701	3.594	.300	.070	.186	--	.063	—	—	—	--	-A14
2.791	3.875	.290	.070	.188	--	.063	—	-141	-142	-143	--
2.913	3.911	.300	.070	.188	--	.063	—	--	--	--	-A15
2.972	4.109	.290	.070	.188	--	.072	--	-151	-152	-153	—
3.177	4.141	.360	.070	.168	--	.053	--	--	--	--	-A16
3.178	4.375	.359	.070	.185	--	.072	--	-161	-162	-163	--
3.356	4.359	.300	.070	.168	--	.360	--	--	--	--	-A17
3.356	4.625	.359	.070	.188	--	.072	—	-171	-172	-173	--
3.522	4.547	.360	.070	.188	--	.063	—	--	--	--	-A18
3.522	4.938	.353	.070	.168	--	.094	—	-181	-182	-183	--
3.800	4.812	.360	.070	.188	--	.063	—	—	—	—	-A19
3.600	5.219	.353	.070	.188	—	.094	—	-191	-192	-193	—
3.988	5.500	.360	.085	.250	--	.063	—	--	—	—	-A20
3.986	5.000	.353	.085	.250	—	.094	--	-201	-202	-203	--
4.192	5.188	.360	.085	.250	--	.063	--	—	—	--	-A21
4.192	5.703	.353	.085	.250	—	.094	--	-211	-212	-213	—
4.395	5.406	.360	.085	.250	—	.063	—	—	—	--	-A22
4.355	6.062	.353	.085	.250	--	.125	—	-221	-222	-223	—
4.801	6.469	.353	.085	.250	--	.125	—	-241	-242	-243	—
5.191	7.031	.435	.105	.250	--	.125	—	-261	-262	-263	—
5.582	7.438	.920	.105	.250	--	.125	—	-281	-282	-283	—
5.983	8.062	.590	.105	.312	--	.156	—	-301	-302	-303	--
6.389	8.438	.590	.105	.312	--	.156	—	-321	-322	-323	—
6.764	9.062	.715	.105	.312	—	.156	—	-341	-342	-343	—
7.171	9.438	.715	.105	.312	—	.156	--	-361	-362	-363	—
7.577	9.875	.715	.105	.312	--	.156	—	-381	-382	-383	—
7.982	10.312	.840	.105	.312	—	.156	—	-401	-402	-403	—

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 402
WASHER, KEY, CUP, LOCK
APPLICABLE DOCUMENT MS9080



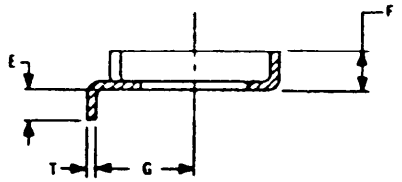


TABLE I Part numbers.

Material	CRES
Protective finish	None

B Min	A Max	C Max	E Min	F Max	G Min	T Max	H	MS9080 Dash no.
.230	.110	.377	.095	.140	.226	.025	.347	-09
.290	.110	.440	.100	.165	.270	.036	.410	-10
.352	.140	.502	.100	.165	.330	.036	.472	-11
.415	.140	.564	.140	.165	.370	.036	.534	-12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 403
WASHER, KEY, CUP, LOCK, SPANNER NUT
APPLICABLE DOCUMENT MS9952

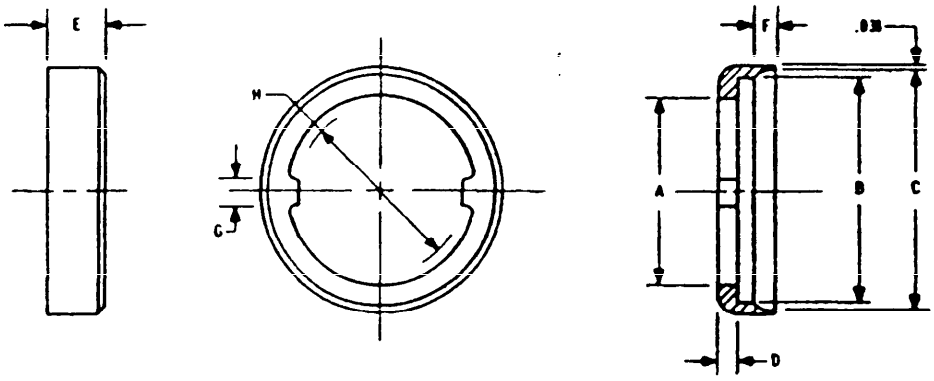


TABLE I. Part numbers.

Material								CRES
Protective finish								None
Hardness (BHN)								140-220
A Min	B Min	C	D	E	F	G Max	H Min	MS9952 + Dash no.
.378	.600	.615	.125	.425	.200	.120	.2756	-02
.441	.750	.775	.125	.425	.200	.120	.3346	-03
.503	.800	.825	.125	.425	.200	.120	.3906	-04
.628	.925	.950	.125	.425	.200	.120	.5156	-06
.753	1.075	1.100	.125	.425	.200	.120	.6406	-08
.878	1.200	1.225	.125	.425	.200	.150	.7656	-10
1.003	1.350	1.375	.125	.425	.200	.150	.8906	-12
1.128	1.475	1.500	.150	.475	.225	.150	1.0156	-14
1.253	1.675	1.650	.150	.475	.225	.180	1.1130	-16
1.378	1.750	1.775	.150	.475	.225	.180	1.2500	-18
1.503	1.900	1.925	.150	.475	.225	.180	1.375	-20
1.753	2.175	2.200	.150	.475	.225	.245	1.630	-24
2.003	2.450	2.475	.150	.475	.225	.245	1.865	-28
2.253	2.700	2.750	.150	.525	.275	.245	2.115	-32
2.503	2.975	3.025	.150	.525	.275	.245	2.365	-36
2.753	3.250	3.300	.150	.525	.275	.245	2.590	-39
3.003	3.525	3.575	.150	.525	.275	.245	2.835	-41

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 404
WASHER, KEY, DRIVE (SYNCHRO)
APPLICABLE DOCUMENT: MS17186

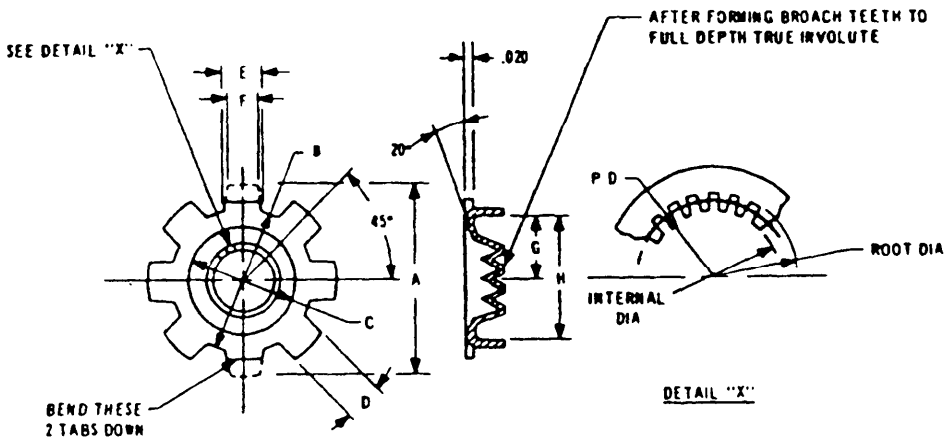


TABLE I. Part numbers.

Material														Aluminum alloy	Brass
Protective finish														Dichromate	Black nickel plating
C Min	A	B	D Min	E	F	G	H	Gear data: Press angle	Gear data: Number of teeth	Gear data: D. P.	Gear data: P. D. Min	Gear data: Root Min	Gear data: Internal Min	MS17186 - Dash no.	
.215	.432	.287	.09	.093	.078	.146	.292	20°	13	120	.108	.125	.091	-1	-5
.291	.516	.350	.09	.125	.093	.177	.355	20°	15	96	.156	.177	.135	-3	-7
									21	120	.175	.192	.158	-2	-6
.350	.625	.437	.12	.125	.093	.208	.417	20°	22	96	.229	.250	.208	-4	-8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 405
WASHER, KEY, FLAT, GENERAL PURPOSE
APPLICABLE DOCUMENT MS27111

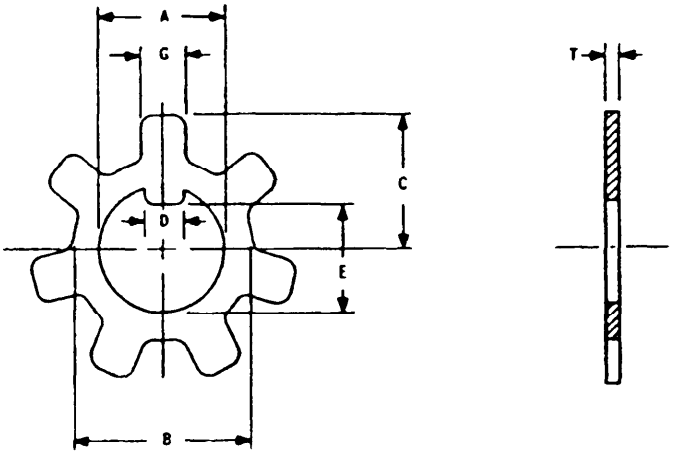


TABLE I. Part numbers.

Material							Carbon steel
Protective finish							Cadmium plate
A	B	C	D	E	G	T Max	MS27111 + Dash no.
.256	.374	.317	.055	.223	.094	.033	-1
.319	.436	.364	.055	.286	.116	.033	-2
.381	.499	.411	.086	.333	.140	.033	-3
.444	.560	.458	.086	.396	.164	.033	-4
.506	.684	.546	.117	.442	.188	.033	-5
.631	.896	.705	.148	.551	.234	.052	-7
.756	1.019	.798	.180	.661	.280	.052	-8
.881	1.206	.934	.180	.786	.328	.052	-9
1.006	1.367	1.052	.242	.880	.374	.065	-10
1.131	1.517	1.211	.242	1.005	.422	.065	-11
1.256	1.698	1.361	.305	1.097	.468	.065	-12
1.381	1.880	1.487	.305	1.222	.516	.065	-13
1.506	2.029	1.623	.367	1.316	.562	.081	-14
1.756	2.392	1.875	.430	1.534	.656	.081	-15
2.006	2.754	2.127	.492	1.753	.750	.081	-16

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 406
WASHER, KEY, LOCKING
APPLICABLE DOCUMENTS MS9276, MS9581, MS9582

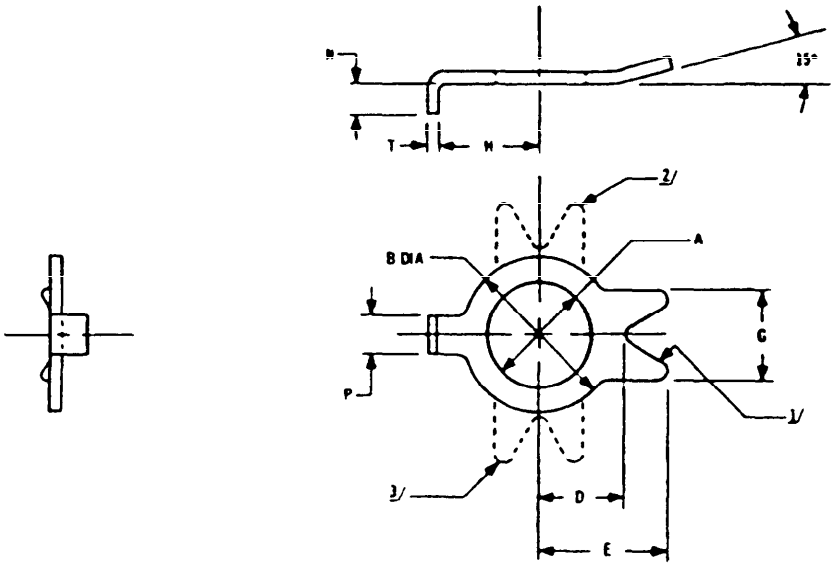


TABLE I. Part numbers.

Material									CRES		
Protective finish									Not specified	Not specified	Not specified
Configuration									1/	2/	3/
Angle									180°	90°	270°
A Min	B Max	D Min	E Min	G Max	H Max	N Min	P Max	T Max	MS9276 - Dash no.	MS9581 - Dash no.	MS9582 - Dash no.
.170	.320	.160	.275	.185	.191	.080	.085	.025	-08	-08	-04
.195	.380	.190	.295	.210	.231	.095	.115	.025	-09	-09	-09
.255	.450	.220	.350	.260	.275	.100	.115	.036	-10	-10	-10
.320	.505	.250	.410	.290	.335	.100	.145	.036	-11	-11	-11
.390	.570	.280	.450	.330	.375	.140	.145	.036	-12	-12	-12
.450	.695	.345	.525	.400	.425	.150	.175	.040	-13	-13	-13
.515	.755	.375	.590	.435	.475	.150	.205	.040	-14	-14	-14
.640	.940	.470	.680	.540	.575	.150	.205	.040	-16	-16	-16
.765	1.065	.530	.730	.560	.675	.150	.235	.040	-17	-17	-17
.890	1.255	.625	.860	.610	.800	.150	.265	.040	-18	-18	-18
1.015	1.440	.715	.950	.660	.875	.150	.295	.040	-19	-19	-19

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 407
WASHER, KEY, LOCKING, SPECIAL DESIGN
APPLICABLE DOCUMENT: MS15820

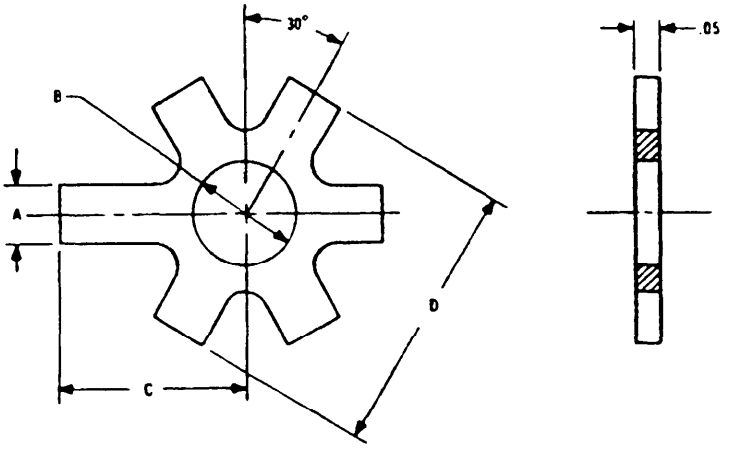


TABLE I. Part numbers.

Material				Steel
Protective finish				Galvanized
B	A	C	D	MS15820 - Dash no.
13/32	1/4	1	1-1/4	-1
17/32	5/16	1-1/8	1-1/2	-2
21/32	3/8	1-1/4	1-3/4	-3
25/32	3/8	1-3/8	2	-4
29/32	5/16	1-1/2	2-1/4	-5
1-1/16	1/2	1-3/4	2-1/2	-6
1-3/16	1/2	1-3/4	2-1/2	-7
1-5/16	1/2	1-7/8	2-5/8	-8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 408
WASHER, KEY, RETAINING, WHEEL BEARING
APPLICABLE DOCUMENT MS21258

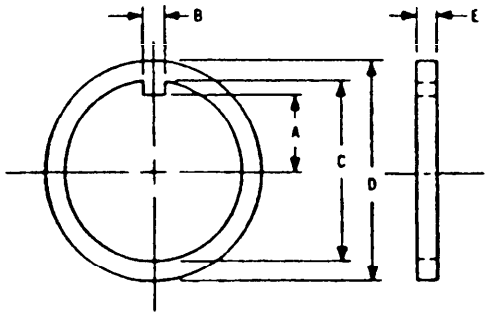


TABLE I. Part numbers.

Material					Steel	
Protective finish					Cadmium plate	Zinc plate
A Min	B Max	C Min	D Max	E Max	MS21258 - Dash no.	
.388	.135	.944	1.405	.098	C15	Z15
.419	.166	1.005	1.530	.130	C16	Z16
.541	.166	1.255	1.780	.130	C20	Z20
.630	.166	1.440	1.905	.130	C23	Z23
.665	.166	1.510	1.905	.130	C24	Z24
.750	.166	1.755	2.280	.130	C28	Z28
.845	.229	1.945	2.405	.130	C31	Z31
.875	.229	2.005	2.655	.130	C32	Z32
1.100	.229	2.445	3.092	.130	C39	Z39
1.302	.291	2.943	3.718	.130	C47	Z47
1.552	.291	3.443	4.780	.130	C55	Z55
2.052	.291	4.445	5.405	.192	C71	Z71

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 409
WASHER, KEY, TAB
APPLICABLE DOCUMENTS MS25081, NAS1636

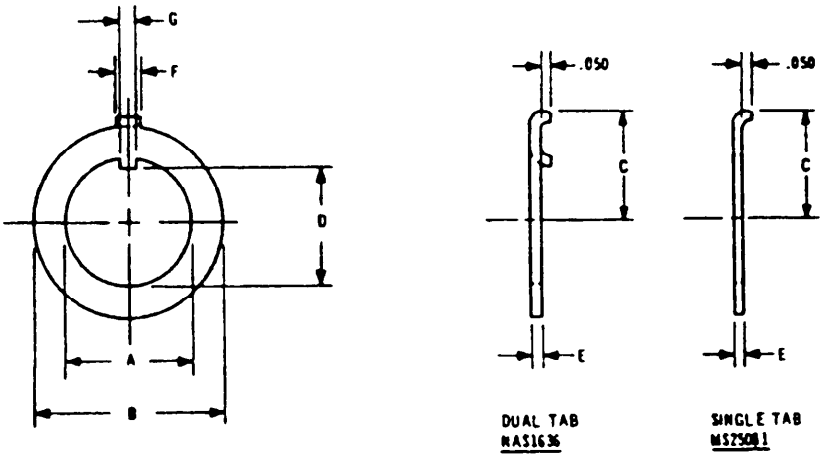


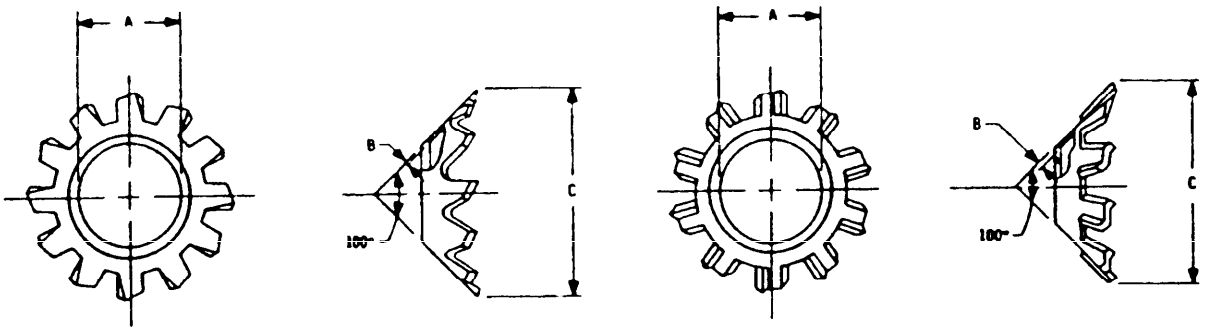
TABLE I. Part numbers.

Material							Carbon steel	CRES	
Protective finish							Cadmium plate	Passivate	
A Min	B Max	C Max	D Min	E Max	F Max	G Max	MS25081 - Dash no.		NAS1636 - Dash no.
.250	.414	.268	.220	.035	.067	.051	-0	C0	—
.377	.642	.370	.340	.035	.115	.065	—	—	-3
.470	.735	.397	.433	.045	.115	.067	-4	C4	—
.470	.735	.417	.433	.035	.115	.065	—	—	-1
.626	.891	.522	.589	.045	.115	.067	-5	C5	—
.626	.891	.494	.589	.035	.115	.065	—	—	-2
.647	.891	.475	.581	.045	.115	.067	-6	C6	—
.747	1.000	.533	.693	.045	.115	.067	-7	C7	—
1.005	1.266	.710	.951	.045	.115	.067	-8	C8	—

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 501
WASHER, LOCK, EXTERNAL TOOTH, COUNTERSUNK
APPLICABLE DOCUMENT MS35790



OPTIONAL DESIGNS

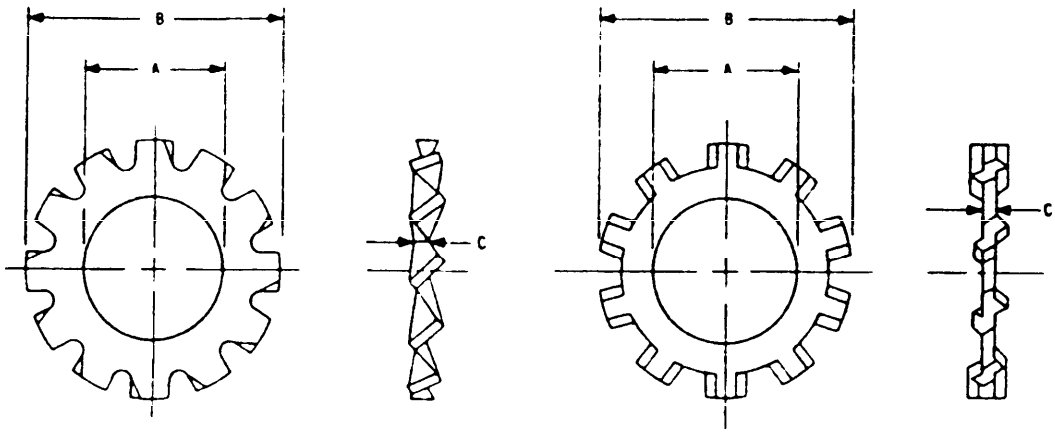
TABLE I. Part numbers.

Material			Carbon steel	Tin brass or phosphor bronze
Protective finish			Cadmium plate	None
A Min	B Max	C Max	MS35790 + Dash no.	
.089	.010	.193	-101	-117
.113	.019	.280	-109	-119
.140	.021	.319	-1	-121
.167	.021	.360	-9	-123
.195	.025	.394	-17	-125
.255	.025	.493	-25	-127
.318	.028	.653	-33	-129
.383	.034	.850	-41	-131

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 502
WASHER LOCK, EXTERNAL TOOTH, FLAT
APPLICABLE DOCUMENT MS35335



OPTIONAL DESIGNS

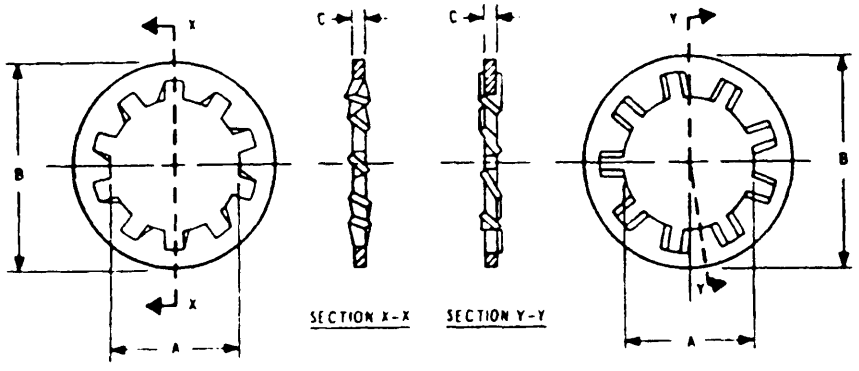
* TABLE I Part numbers.

Material			Carbon steel	CRES	Tin, brass or phosphor bronze
Protective finish			Cadmium plate	Passivate	Uncoated
A Min	B Max	C Max	MS35335 - Dash no.		
.115	.260	.019	-29	-57	-85
.141	.320	.022	-30	-58	-86
.168	.381	.023	-31	-59	-87
.195	.410	.025	-32	-60	-88
.256	.510	.028	-33	-61	-89
.320	.610	.034	-34	-62	-90
.384	.694	.040	-35	-63	-91
.448	.760	.040	-36	-64	-92
.513	.900	.045	-37	-65	-93
.576	.985	.045	-38	-66	-94
.641	1.070	.050	-39	-67	-95
.768	1.260	.055	-40	-68	-96
.897	1.410	.060	-41	-69	-97
1.025	1.620	.067	-42	-70	-98

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 503
WASHER, LOCK, INTERNAL TOOTH, FLAT
APPLICABLE DOCUMENTS MS35333, MS35334



OPTIONAL DESIGNS

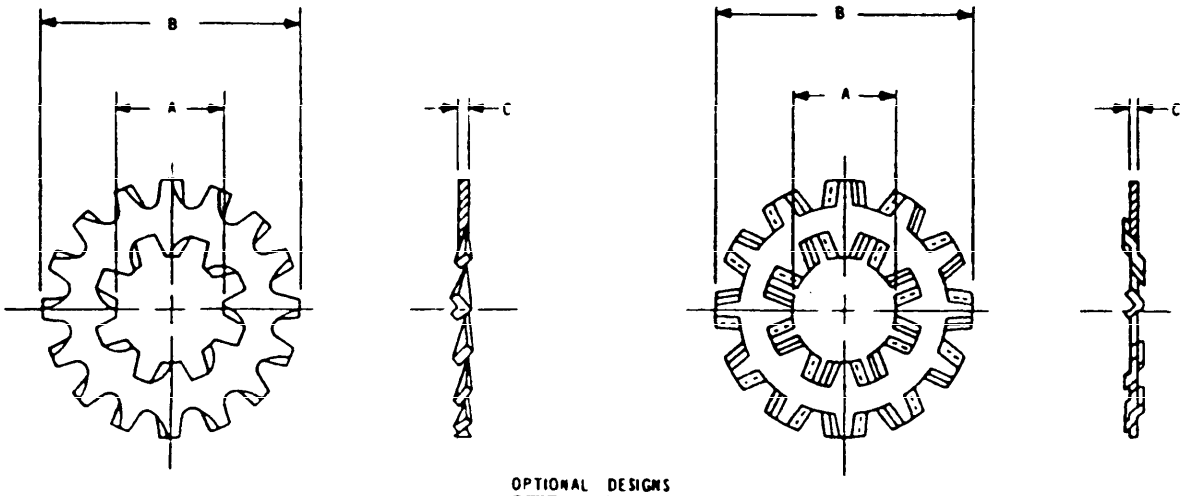
TABLE I Part numbers

Material			Carbon steel		CRES	Tin brass or phosphor bronze
Protective finish			Cadmium plate		Passivate	None
A Min	B Max	C Max	MS35333 - Dash no.	MS35334 - Dash no.	MS35333 - Dash no.	
.089	.200	.015	-35	—	-69	-103
.115	.270	.019	-36	--	-70	-104
.141	.295	.021	-37	—	-71	-105
.168	.340	.023	-38	--	-72	-106
.195	.381	.025	-39	--	-73	-107
.256	.410	.020	-125	--	-135	-145
	.478	.028	-40	—	-74	-108
	.536	.045	--	-19	--	—
.320	.607	.050	—	-20	—	—
	.610	.034	-41	--	-75	-109
.384	.692	.040	-42	--	-76	-110
	.748	.050	--	-21	--	--
.440	.592	.020	-131	--	-141	-151
.448	.789	.040	-43	--	-77	-111
.512	.900	.045	-44	—	-78	-113
	.924	.062	--	-23	--	—
.630	.800	.027	-128	--	-138	-148
.640	1.071	.050	-46	—	-80	-115
	1.135	.062	--	-25	—	--
.768	1.265	.084	--	-26	--	--
.769	1.245	.055	-47	--	-81	-116
.775	1.077	.027	129	--	-139	-149
.894	1.410	.060	-48	--	-82	-117
	1.447	.084	--	-27	--	—
1.010	1.390	.028	-130	--	-140	-150
1.019	1.637	.062	-49	--	-83	-118
1.275	1.975	.062	-51	--	-85	-120

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 504
WASHER LOCK INTERNAL AND EXTERNAL TOOTH, FLAT
APPLICABLE DOCUMENT MS45904



OPTIONAL DESIGNS

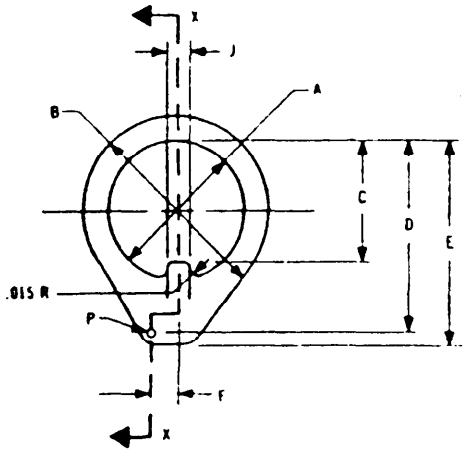
TABLE I. Part numbers.

Material			Carbon steel
Protective finish.			Cadmium plate
A Min	B Max	C Max	MS45904 + Dash no.
115	.475	.021	-51
.141	.510	.028	-54
.168	.610	.034	-57
	.690	.034	-58
	.760	.034	-59
.195	.610	.034	-60
	.690	.040	-61
	.760	.040	-62
	.900	.040	-63
.256	.760	.040	-68
	.900	.040	-69
	.985	.045	-70
	1.070	.045	-71
.320	.900	.040	-72
	.985	.045	-73
	1.070	.050	-74
.384	.985	.045	-76
	1.070	.050	-77
	1.155	.050	-78
	1.260	.050	-79
.512	1.260	.055	-84
	1.410	.060	-86
	1.620	.067	-87
.640	1.410	.060	-92
	1.620	.067	-93

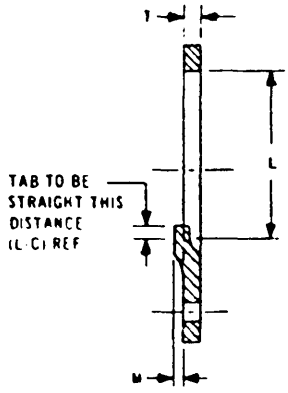
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

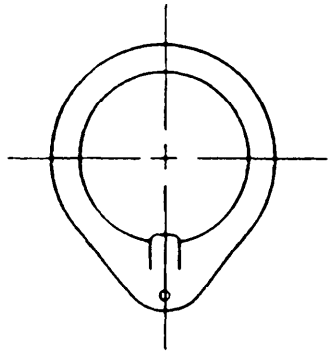
SECTION 505
WASHER, LOCK, ROD END
APPLICABLE DOCUMENT: NAS513



-4 THRU -20



SECTION X-X



-22 THRU -36

TABLE I. Part numbers.

Material											Spring steel
Protective finish											Cadmium plate
Hardness (Rockwell)											C40-45
A	B	C	D	E	F	J	L	M	T	P	NAS513 Dash no.
.272	.44	.214	.406	.500	.125	.052	.293	.036	.050	.062	-4
.334	.50	.273	.484	.578	.125	.052	.355	.036	.050	.062	-5
.396	.56	.327	.562	.656	.125	.082	.418	.036	.050	.062	-6
.459	.63	.366	.625	.719	.125	.082	.463	.049	.063	.062	-7
.520	.75	.450	.719	.813	.156	.082	.542	.049	.063	.062	-8
.647	1.00	.559	.937	1.031	.188	.114	.675	.057	.071	.071	-10
.772	1.12	.681	1.062	1.156	.188	.114	.800	.057	.071	.071	-12
.897	1.31	.795	1.250	1.344	.188	.142	.944	.066	.080	.089	-14
1.022	1.50	.918	1.422	1.516	.188	.142	1.088	.074	.090	.090	-16
1.147	1.62	1.028	1.547	1.656	.219	.174	1.213	.074	.090	.090	-18
1.272	1.75	1.154	1.687	1.796	.219	174	1.280	.096	.112	.112	-20
1.397	1.88	1.254	1.859	1.969	--	.236	1.405	.096	112	.112	-22
1.518	2.00	1.379	1.969	2.078	--	.236	1.530	.096	.112	.112	-24
1.643	2.12	1.495	2.109	2.219	--	.236	1.638	.109	.125	.125	-26
1.766	2.25	1.607	2.250	2.375	--	.298	1.763	.109	.125	.125	-28
1.891	2.38	1.732	2.375	2.484	--	.298	1.888	.109	.125	.125	-30
2.016	2.63	1.857	2.594	2.703	--	.298	2.076	.109	.125	.125	-32
2.141	2.75	1.973	2.750	2.860	--	.298	2.201	.109	.125	.125	-34
2.266	2.88	2.098	2.875	2.984	--	.298	2.326	.109	.125	.125	-36

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 506
WASHER LOCK SPRING HELICAL
APPLICABLE DOCUMENTS MS35338, MS35340, MS51848

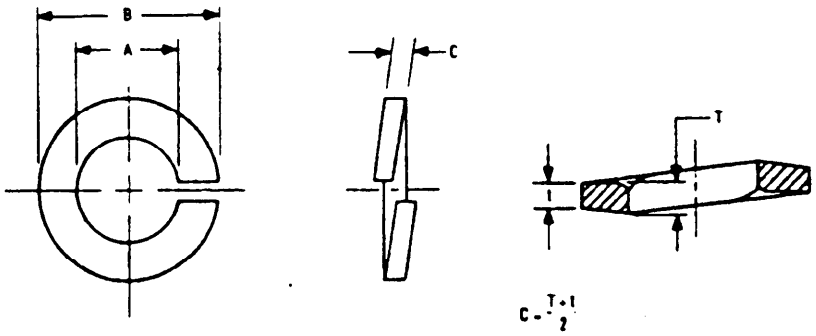


TABLE I. Part numbers.

Material			Carbon steel					CRES Series					Ph brz or tin brs	Nickel copper alloy
								300	300	420	300	420		
Protective finish			Cadmium plate			Phosphate		Passivate			Black oxide		Cadmium plate	None
Hardness (Rockwell)			C45-43	C45-53	C45-53	Not specified		C35-43	C35-43	C45-53	C35-43	C45-53	None	
A Min	B Max	C Min	MS51848 - Dash no.	MS35338 - Dash no.	MS35340 - Dash no.	MS35338 - Dash no.	MS51848 - Dash no.	MS51848 - Dash no.	MS35338 - Dash no.					
.062	.092	.016	-1	—	—	—	-21	-41	—	—	—	—	--	—
.088	.135	.020	-3	—	—	—	-23	-43	—	—	—	—	—	—
	.172	.020	--	-39	—	-58	—	—	-134	-153	-1348	-1538	-96	-115
	.208	.027	—	--	-39	—	—	—	—	—	—	—	—	—
.114	.173	.022	-5	--	—	—	-25	-45	—	—	—	—	—	—
	.115	.034	—	—	-40	—	—	—	—	—	—	—	—	—
.115	.209	.025	--	-40	--	-59	--	—	-135	-154	-1358	-1548	-97	-116
.141	.216	.030	-7	--	—	—	-27	-47	—	—	—	—	--	—
	.250	.031	—	-41	--	-60	—	—	-136	-155	-1368	-1558	-98	-117
	.314	.045	—	—	-41	—	—	—	—	—	—	--	—	—
.167	.267	.047	-8	—	--	—	-28	-48	—	—	—	—	—	—
	.168	.057	--	--	-42	—	—	—	—	—	—	—	--	—
.168	.293	.040	—	-42	—	-61	—	—	-137	-156	-1378	-1568	-99	-118
.193	.294	.047	-9	—	--	—	-29	-49	—	—	--	—	—	—
	.194	.068	--	—	-43	--	—	—	—	--	—	--	--	—
.194	.334	.047	—	-43	—	-62	—	—	-138	-157	-1388	-1578	-100	-119
.254	.365	.078	-10	--	--	--	-30	-50	—	—	—	—	--	--
	.255	.084	—	--	-44	—	—	--	--	—	—	--	--	—
.255	.489	.062	—	-44	--	-63	--	—	-139	-158	-1398	-1588	-101	-120
.317	.460	.093	-11	--	--	--	-31	-51	—	—	--	--	--	--
	.318	.108	--	--	-45	--	--	--	—	--	—	--	--	--

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

TABLE I Part numbers. - Continued

Material			Carbon steel					CRES					Phosphor or tin bronze	Nickel copper alloy
								Series						
								300	300	420	300	420		
Protective finish			Cadmium plate			Phosphate		Passivate			Black oxide		Cadmium plate	None
Hardness (Rockwell)			C45-53	C45-53	C45-53	C45-53		C35-43	C35-43	C45-53	C35-43	C45-53	None	
A Min	B Max	C Min	MS51848 - Dash no.	MS35338 - Dash no.	MS35340 - Dash no.	MS35338 - Dash no.	MS51848 - Dash no.	MS51848 - Dash no.	MS35338 - Dash no.					
.318	.586	.078	--	-45	—	-64	—	--	-140	-159	-140B	-159B	-102	-121
.382	.553 .741	.125 .123	-12 —	— --	-- -46	— --	-32 --	-52 —	-- --	-- --	-- --	— --	-- --	-- --
.382	.683	.094	--	-46		-65	--	--	-141	-160	-141B	-160B	-103	-122
.443	.647	.140	-13	--	--	--	-33	-53	—	--	--	--	--	--
.446	.779	.109	--	-47	--	-66	--	—	-142	-161	-142B	-161B	-104	-123
.509	.737 .939	.172 .162	-14 —	— --	— -48	— --	-34 --	-54 —	— —	— —	— --	— --	— --	— --
.509	.873	.125	--	-48	—	-67	—	—	-143	-162	-143B	-162B	-105	-124
.572	.971	.141	--	-49	—	-68	—	—	-144	-163	-144B	-163B	-106	-125
.636	.923 1.157	.203 .202	-15 —	— —	— -50	— —	— —	-55 —	— —	— —	— —	— —	— —	— —
.636	1.079	.156	--	-50	—	-69	—	—	-145	-164	-145B	-164B	-107	-126
.763	1.111 1.361	.218 .241	-16 --	— --	— -51	— —	— —	— —	— —	— —	— --	— --	— —	— —
.763	1.271	.188	--	-51	—	-70	—	—	-146	-165	-146B	-165B	-108	-127
.887	1.296	.234	-17	--	—	—	—	--	—	--	--	--	--	--
.890	1.464	.219	—	-52	--	-71	—	—	-147	-166	-147B	-166B	-109	-128
1.017	1.483 1.661 1.799	.250 .250 .330	-18 -- —	— -53 —	— -- -53	— -72 —	— — —	— — —	— -148 --	— -167 --	— -148B —	— -167B --	— -110 —	— -129 --
1.144	1.853	.281	—	-54	—	--	—	--	-149	-168	-149B	-168B	-111	-130
1.271	2.045 2.231	.312 .417	— --	-55 --	-- -55	— —	— —	— —	-150 —	-169 —	-150B --	-169B —	-112 —	-131 —
1.398	2.239	.344	--	-56	—	—	--	—	-151	-170	-151B	-170B	-113	-132
1.525	2.430 2.638	.375 .496	-- --	-57 —	-- -57	-- --	— --	— —	-152 —	-171 —	-152B —	-171B —	-114 —	-133 —

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T07:52Z
Check the source to verify that this is the currenversion before use.

SECTION 601
WASHER, RECESSED, COUNTERSUNK, ONE SURFACE
APPLICABLE DOCUMENTS: NAS1587, MS20002, MS21206

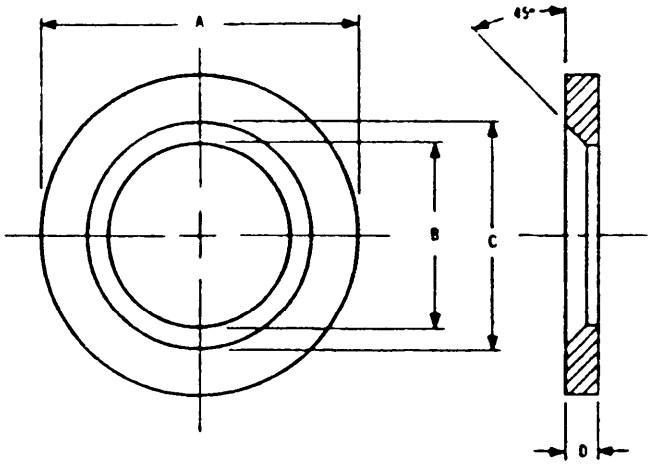


TABLE I. Part numbers.

Material				CRES Type 321 or 347	Alloy steel	
Protective finish				Passivate	Cadmium plate	
Tensile strength (psi)				75,000 Min	125,000 to 145,000	Not specified
Hardness (Rockwell)				Not specified	Not specified	C38-45
B Min	C Min	A Max	D Max	NAS1587 + Dash no.	MS20002 + Dash no.	MS21206 + Dash no.
.191	.280	.372	.069	—	—	C3
.192	.222	.479	.072	-3C	—	—
.251	.340	.470	.069	—	—	C4
.252	.334	.541	.088	-4C	C4	—
.314	.402	.573	.069	—	—	C5
.315	.356	.593	.078	-5C	C5	—
.376	.497	.679	.085	—	—	C6
.378	.483	.697	.088	-6C	C6	—
.439	.559	.783	.085	--	—	C7
.441	.543	.791	.088	-7C	C7	—
.501	.622	.898	.085	—	—	C8
.504	.604	.085	.088	-8C	C8	—
.627	.778	1.125	.085	—	—	C10
.631	.765	1.072	.088	-10C	C10	—
.752	.903	1.350	.085	—	—	C12
.757	.890	1.260	.088	-12C	C12	—
.877	1.028	1.572	.085	--	—	C14
.884	1.015	1.447	.088	-14C	C14	—
1.002	1.153	1.798	.119	--	--	C16
1.010	1.140	1.635	.088	-16C	C16	—
1.127	1.278	2.188	.119	--	—	C18
1.135	1.265	1.885	.088	-18C	C18	—
1.252	1.435	2.260	.119	—	—	C20
1.260	1.427	2.135	.104	-20C	C20	—
1.377	1.560	2.490	.119	--	—	C22
1.385	1.552	2.323	.104	--	C22	—
1.502	1.685	2.723	.119	—	—	C24
1.510	1.677	2.510	.104	—	C24	—

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 602
WASHER, RECESSED, COUNTERSUNK, TWO SURFACES
APPLICABLE DOCUMENTS: MS9482, MS9768, MS14155, MS14177

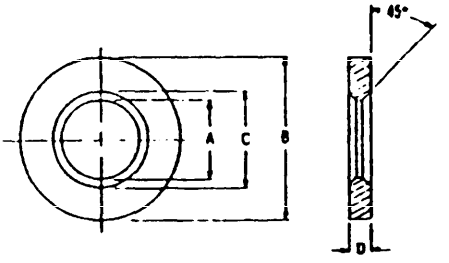


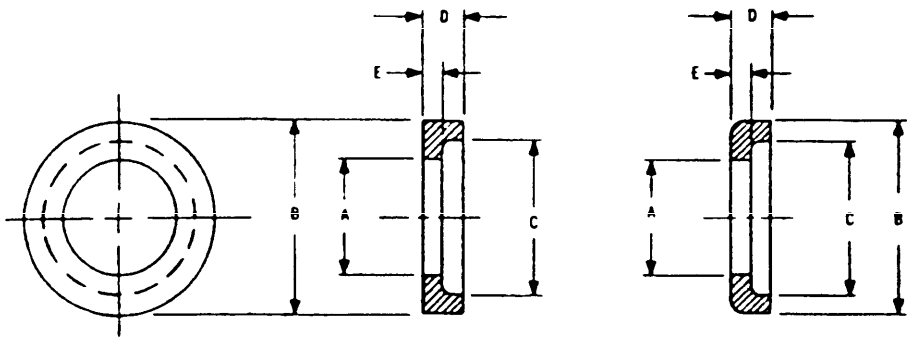
TABLE I. Part numbers.

Material				Steel	CRES	Alloy steel	
Protective finish				Diffused nickel-cadmium plate	None	Cadmium plate	
Hardness (Rockwell)				C50-56	C25-39	C39-43	
A Min	B Max	C Min	D Max	MS9482 • Dash no.	MS9768 • Dash no.	MS14155 • Dash no.	MS14177 • Dash no.
.140	.340	.176	.076	—	-07	—	—
.143	.315	.178	.070	-07	—	—	—
.166	.385	.202	.076	—	-08	—	—
.169	.375	.214	.085	-08	—	—	—
.192	.448	.236	.085	—	-09	—	—
.195	.435	.250	.095	-09	—	—	—
.251	.505	.337	.124	—	—	-4	—
	.532	.337	.133	—	—	—	-4
.252	.635	.296	.085	—	-10	—	—
.256	.570	.332	.135	-10	—	—	—
.314	.620	.399	.124	—	—	-5	—
	.651	.399	.133	—	—	—	-5
.315	.760	.359	.105	—	-11	—	—
.317	.700	.394	.160	-11	—	—	—
.376	.766	.494	.158	—	—	-6	—
	.795	.494	.167	—	—	—	-6
.378	.885	.422	.117	—	-12	—	—
.380	.855	.489	.195	-12	—	—	—
.439	.878	.556	.158	—	—	-7	—
	.912	.556	.167	--	—	—	-7
.441	1.010	.495	.130	—	-13	—	—
.443	.980	.552	.215	-13	—	—	—
.501	1.004	.619	.163	--	—	-8	—
	1.038	.619	.167	—	—	--	-8
.504	1.198	.558	.165	—	-14	—	—
.505	1.170	.614	.260	-14	—	—	—
.627	1.266	.775	.226	—	—	-10	—
	1.302	.775	.198	--	--	—	-10
.630	1.405	.771	.315	-16	—	—	—
.752	1.507	.900	.278	--	—	-12	—
	1.546	.900	.198	--	—	—	-12
.755	1.670	.896	.375	-17	--	—	—
.877	1.747	1.025	.320	—	—	-14	—
	1.791	1.025	.198	—	--	—	-14
.880	1.935	1.021	.430	-18	—	—	—
1.002	1.978	1.150	.368	—	—	-16	—
1.005	2.195	1.146	.490	-19	—	—	—

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 603
WASHER, RECESSED, TAPER PIN
APPLICABLE DOCUMENT AN975



OPTIONAL DESIGN

TABLE I. Part numbers

Material					Steel
Protective finish					Cadmium plate
A	B	C	D	E	AN975 - Dash no.
13/64	15/32	11/32	3/16	1/16	-3
17/64	9/16	13/32	13/64	5/64	-4
21/64	11/16	17/32	13/64	5/64	-5
25/64	25/32	19/32	13/64	5/64	-6
29/64	27/32	21/32	13/64	5/64	-7
33/64	15/16	23/32	7/32	3/32	-8
49/64	1 9/32	1	15/64	7/64	-12
57/64	1 15/32	1 3/16	15/64	7/64	-14

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 701
WASHER SHOULDERED AND RECESSED DIMPLED 100
APPLICABLE DOCUMENT NAS1169

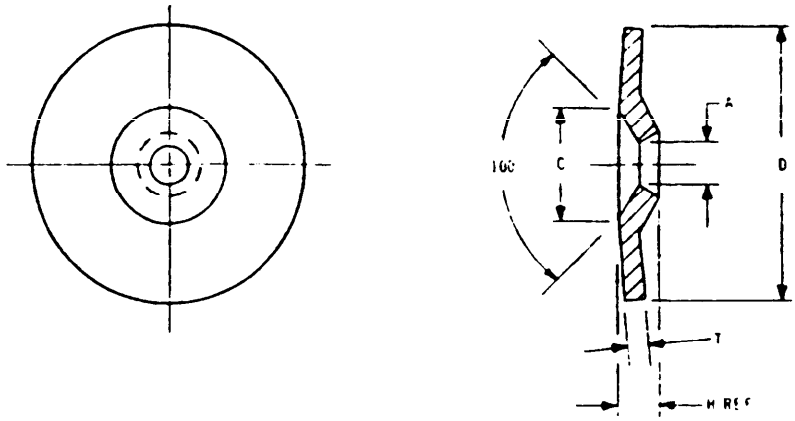


TABLE I Part numbers

Material	Carbon steel	Aluminum alloy		CRES	Titanium
		5052	2024		
Protective finish	Cadmium plate	Anodize		Passivate	Not specified
Hardness or tensile strength (ksi)	Rockwell C42-47	Not specified		160 tensile	None

A Min	D Max	C Min	H Ref	T	NAS1169 - Dash no.				
.140	.495	.279	.068 .076 .082	.008 .016 .025	-- -6L -6	D6E -- --	-- DD6L DD6	-- C6L C6	-- T6L T6
.166	.582	.332	.079 .087 .094	.008 .016 .025	-- -8L -8	D8E -- --	-- DD8L DD8	-- C8L C8	-- T8L T8
192	.645	.385	.091 .099 .101	.008 .016 .025	-- -10L -10	D10E -- --	-- DD10L DD10	-- C10L C10	-- T10L T10
.252	.770	.507	.117 .125 .132	.008 .016 .025	-- -416L -416	D416E -- -	-- DD416L DD416	-- C416L C416	-- T416L T416
.314	.958	.635	.145 .153 .160	.006 .016 .025	-- -516L -516	D516E -- --	-- DD516L DD516	-- C516L C516	-- T516L T516
.377	1.020	.762	.172 .180 .188	.008 .016 .025	-- -616L -616	D616E -- --	-- DD616L DD616	-- C616L C616	-- T616L T616
440	1.208	.890	.200 .208 .215	.008 .016 .025	-- -716L -716	D716E -- --	-- DD716L DD716	-- C716L C716	-- T716L T716
502	1332	1.017	.225 .237 .245	.008 .016 .025	-- -816L -816	D816E -- -	-- DD816L DD816	-- C816L C816	-- T816L T816

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

SECTION 801
WASHER, SPRING TENSION
APPLICABLE DOCUMENT: MIL-W-12133/1

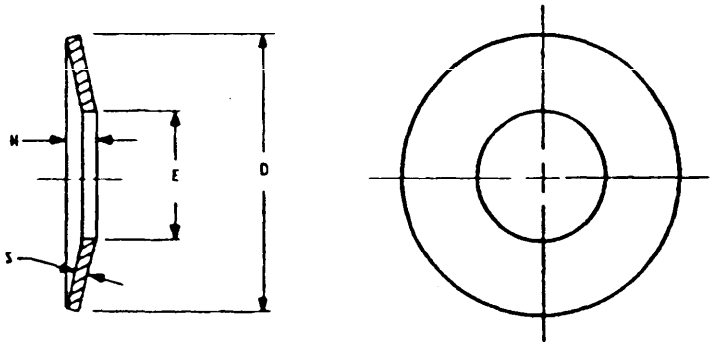


TABLE I. Part numbers.

Material				Spring steel	
Protective finish				Plain	Cadmium plate
E Min	D Max	S Nom	H Nom	MIL-W-12133/1 Dash no.	
.125	.250	.013	.020	-1	-1P
.156	.312	.017	.025	-2	-2P
.195	.375	.015	.027	-3	-3P
.250	.500	.020	.035	-4	-4P
.317	.625	.032	.051	-5	-5P
.382	.750	.032	.054	-6	-6P
.445	1.000	.035	.067	-7	-7P
.512	1.000	.039	.075	-8	-8P
	1.102	.048	.083	-9	-9P
.637	1.375	.059	.102	-10	-10P
.761	1.500	.059	.114	-11	-11P
		.078	.121	-12	-12P
.880	1.750	.085	.128	-13	-13P
1.016	2.000	.079	.138	-14	-14P
	2.375	.098	.177	-15	-15P

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

...tems Division (AFSC)
...n AFB, OH 45433

POSTAGE AND FEES PAID

U.S. MAIL

...SINESS
...TE USE $300

Commander
Aeronautical Systems Division (AFSC)
ATTN: ASD/ENESS
Wright-Patterson AFB, OH 45433

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-STD-1764 WASHERS, PREFERRED FOR DESIGN, LISTING OF

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address – Optional)*

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:52Z
Check the source to verify that this is the current version before use.